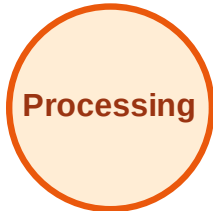


BFS Traversal

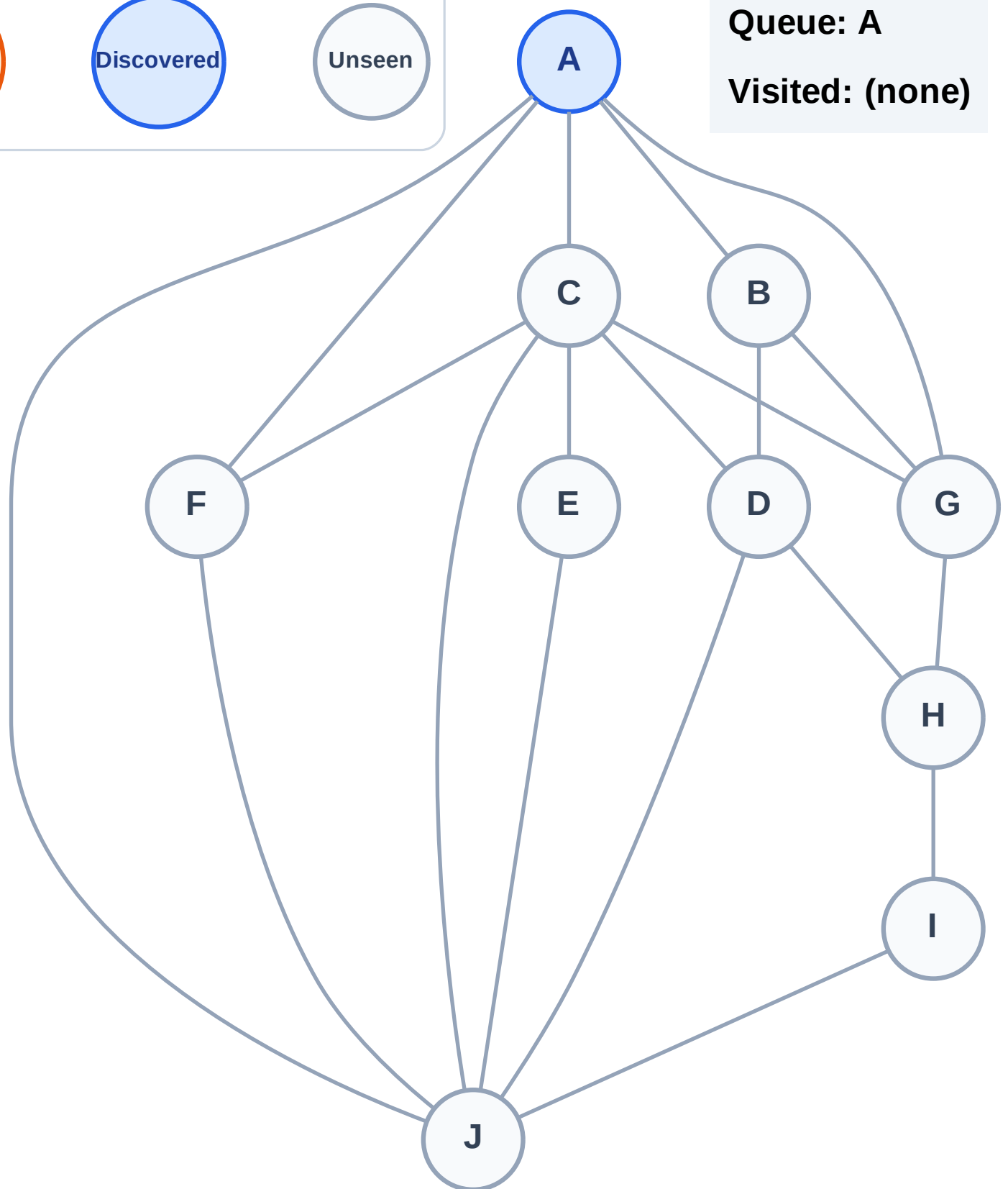
Step 1: Start at A

Legend



Queue: A

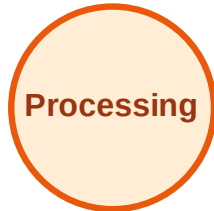
Visited: (none)



BFS Traversal

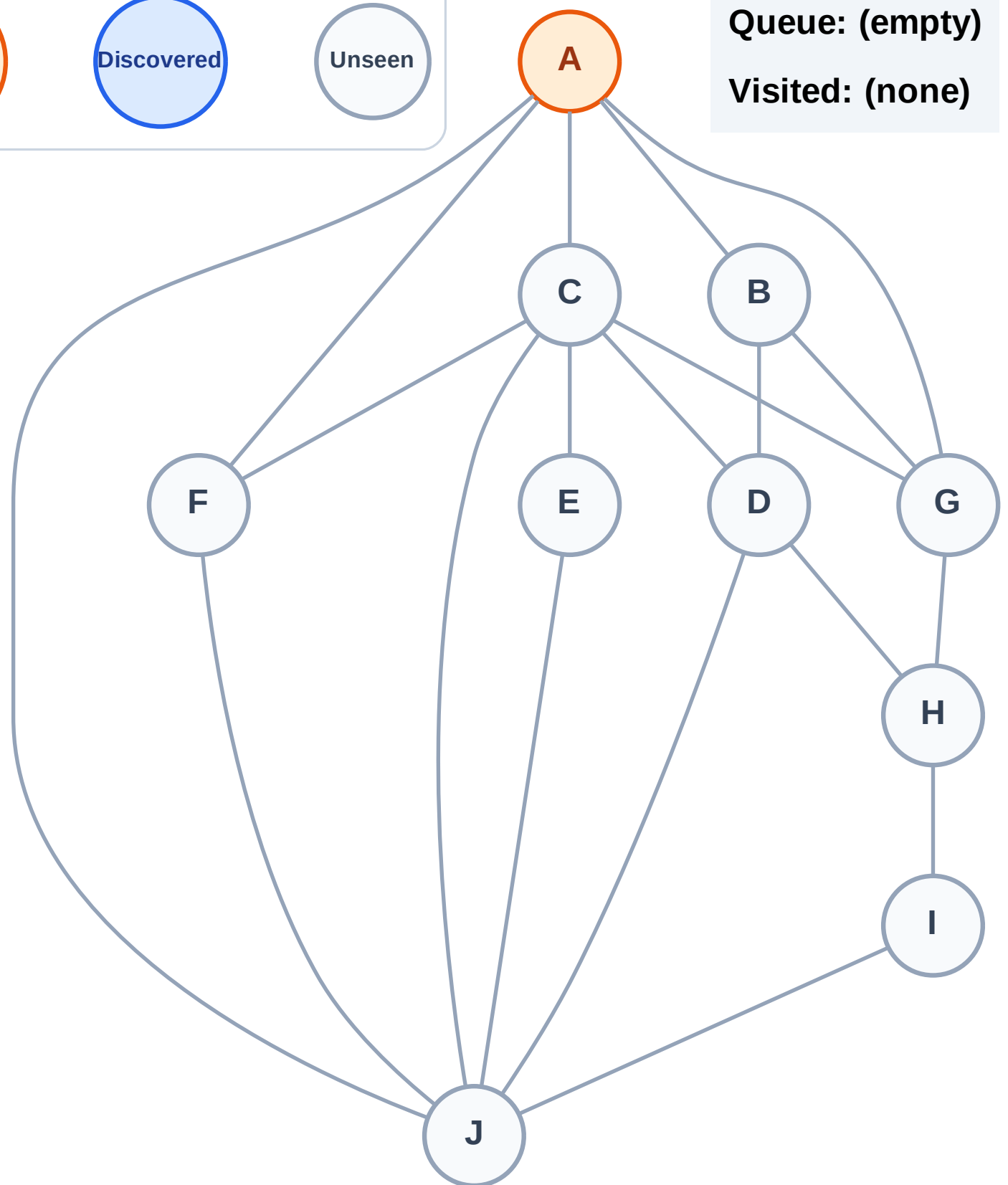
Step 2: Process node A

Legend



Queue: (empty)

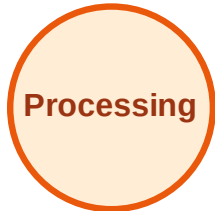
Visited: (none)



BFS Traversal

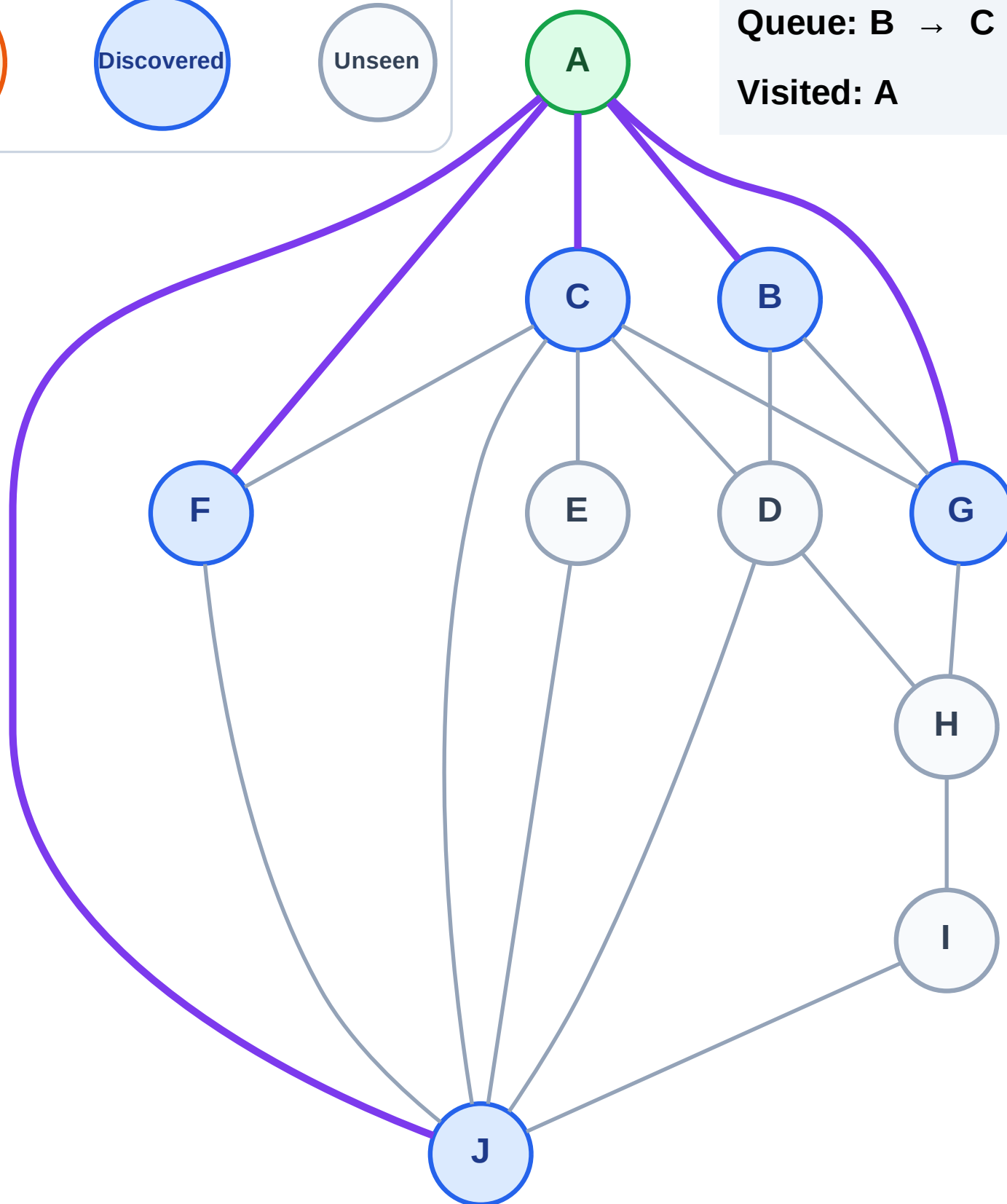
Step 3: Discover B, C, F, G, J from A

Legend



Queue: B → C → F → G → J

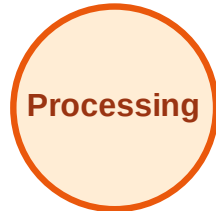
Visited: A



BFS Traversal

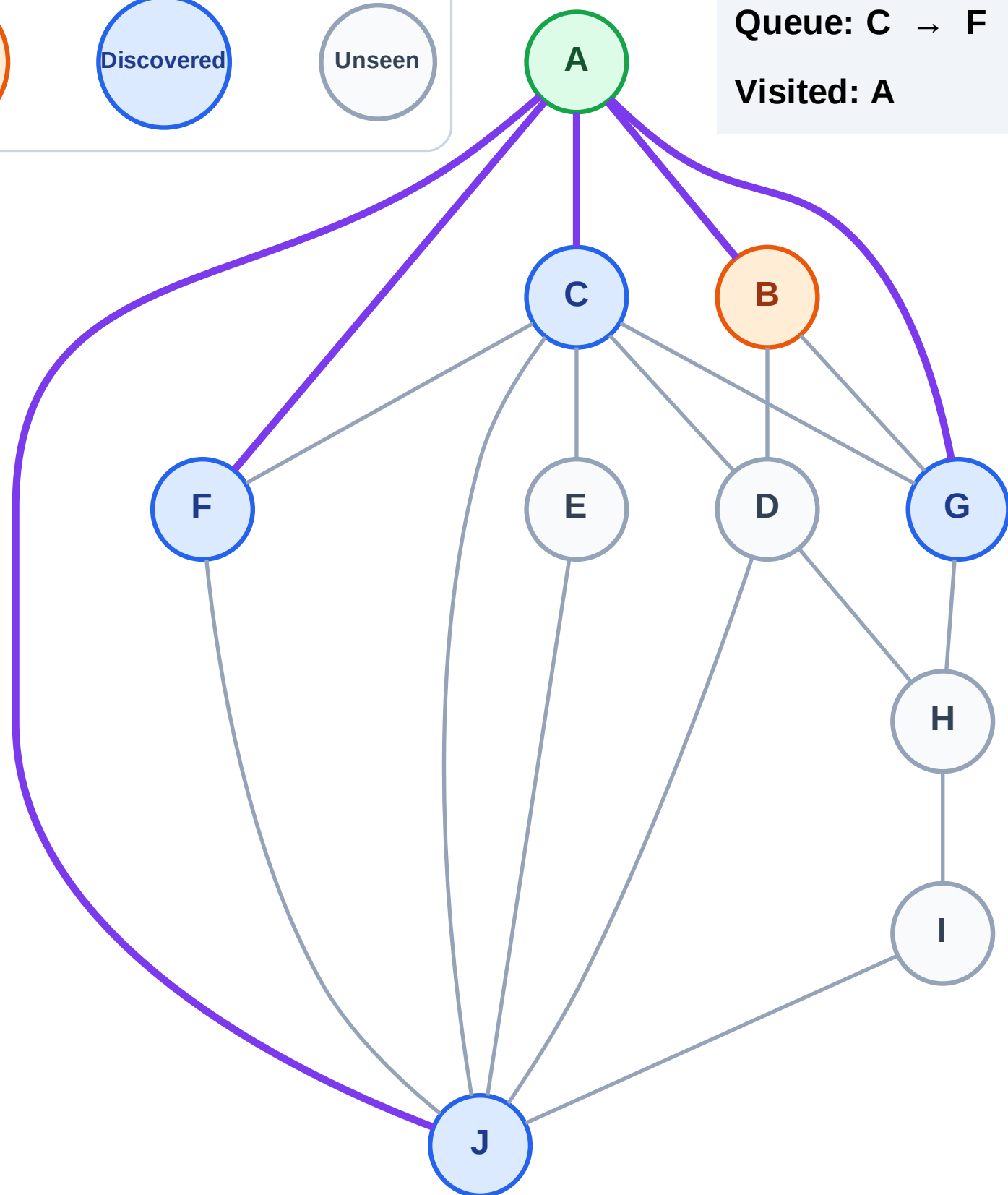
Step 4: Process node B

Legend



Queue: C → F → G → J

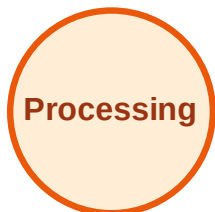
Visited: A



BFS Traversal

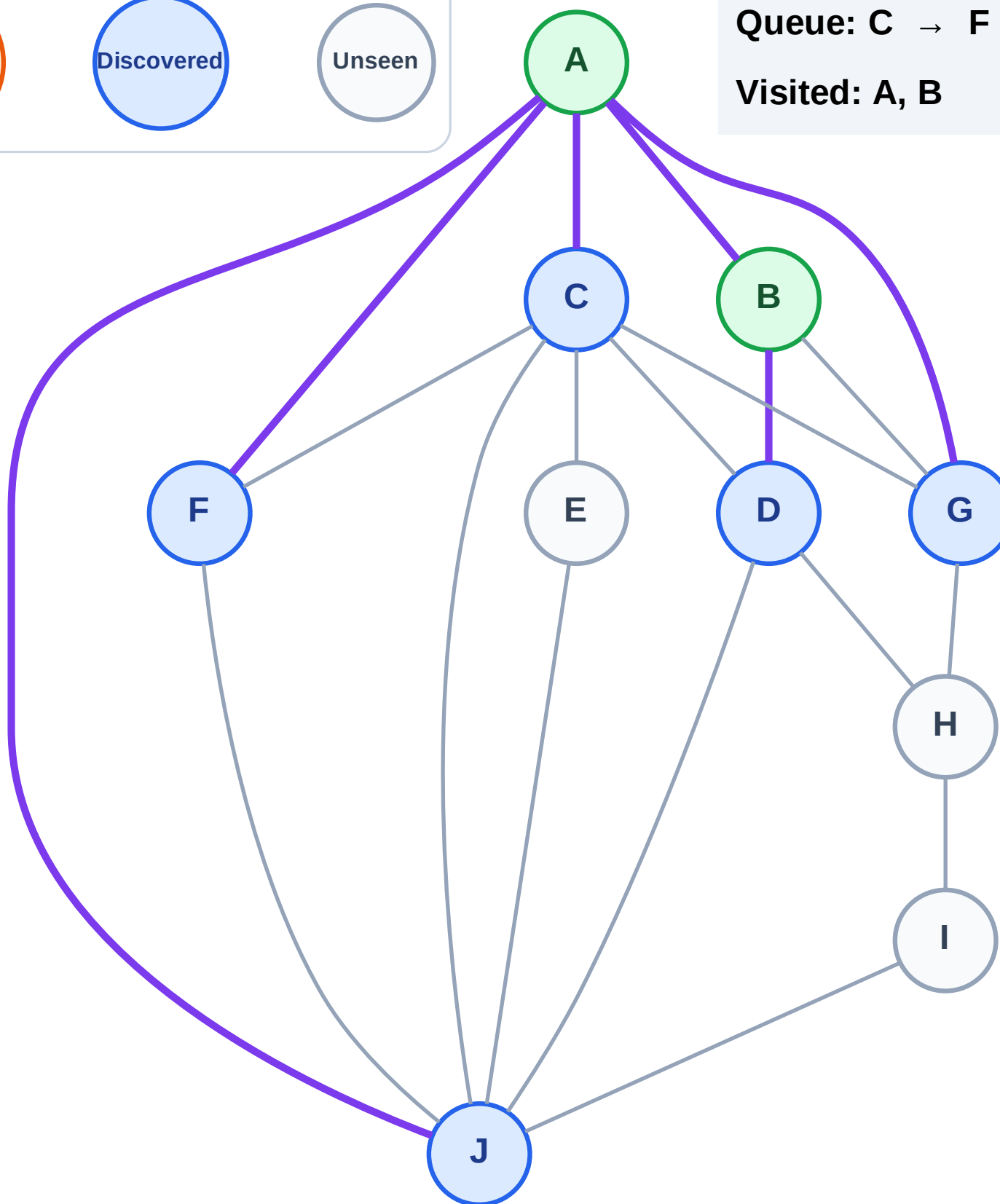
Step 5: Discover D from B

Legend



Queue: C → F → G → J → D

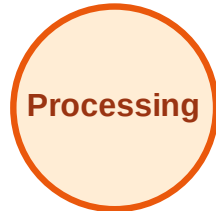
Visited: A, B



BFS Traversal

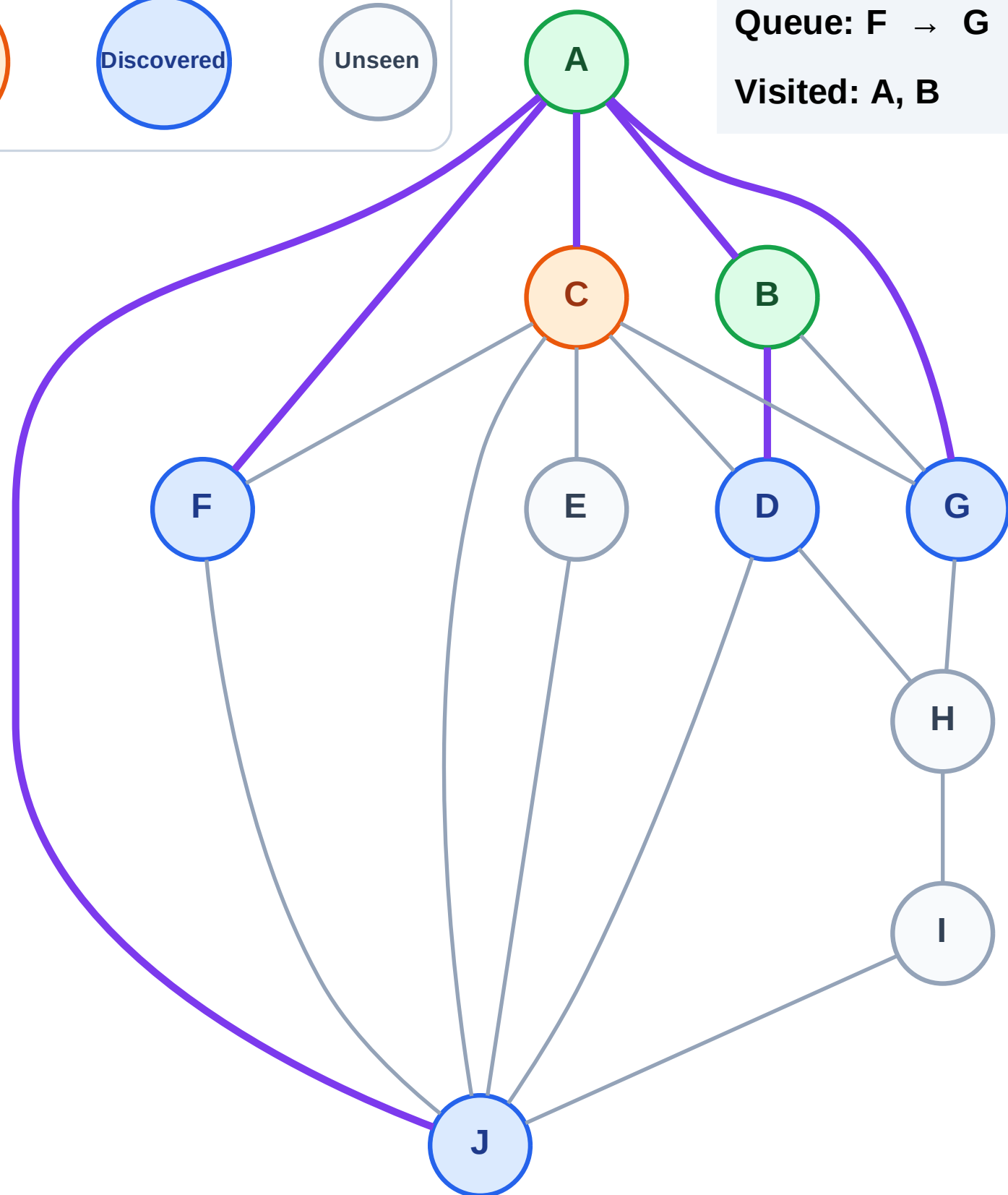
Step 6: Process node C

Legend



Queue: F → G → J → D

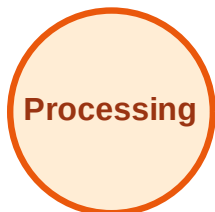
Visited: A, B



BFS Traversal

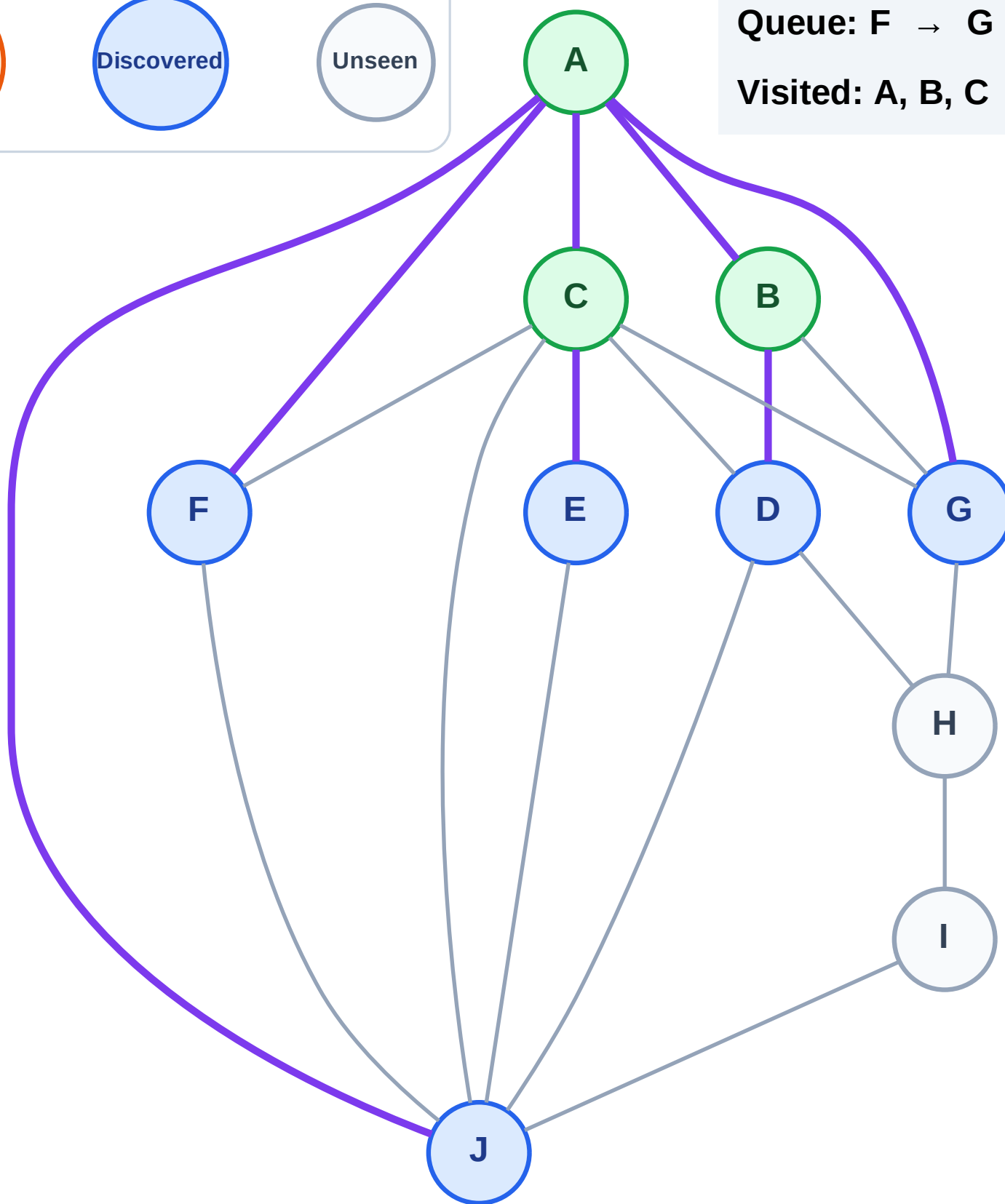
Step 7: Discover E from C

Legend



Queue: F → G → J → D → E

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

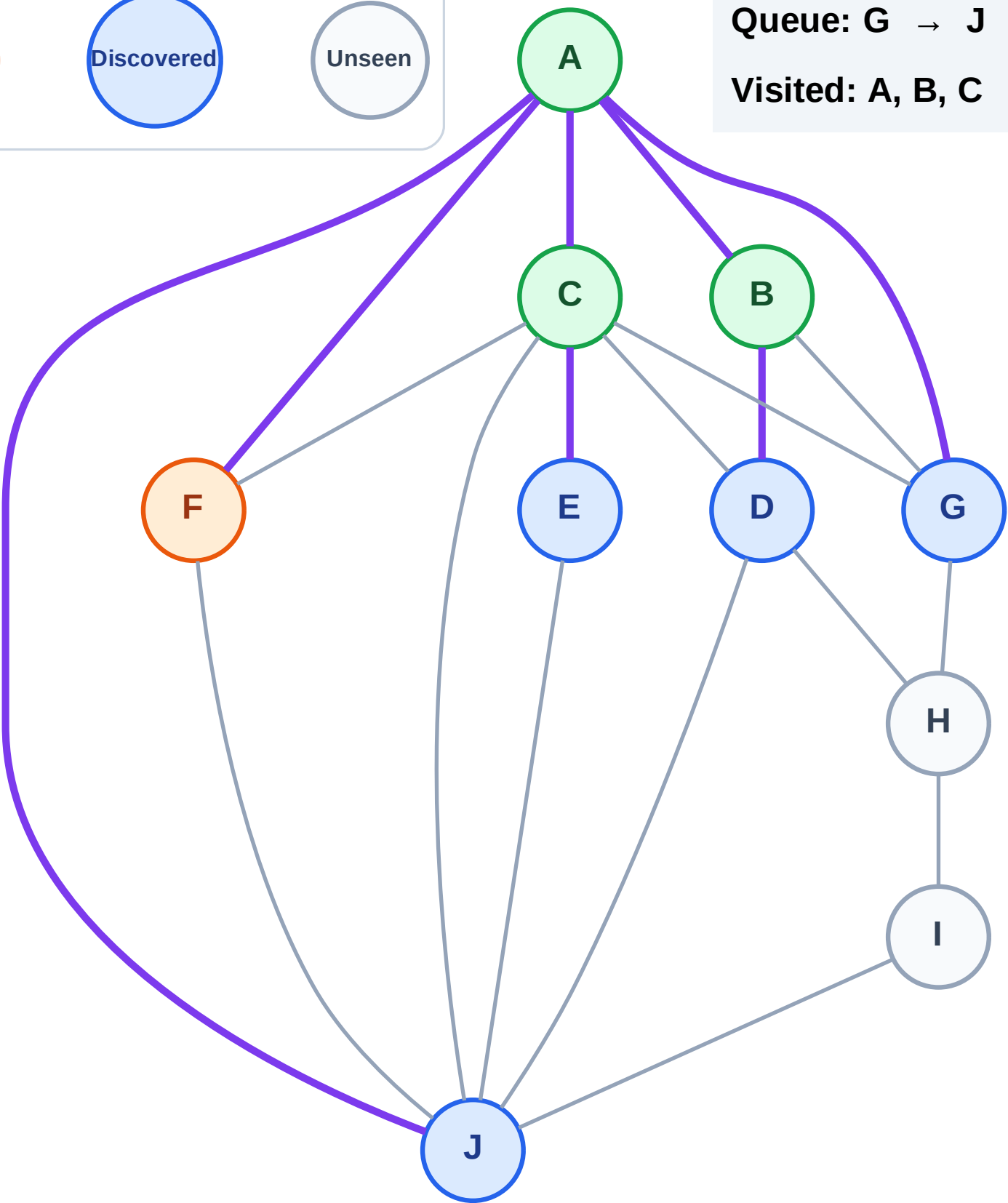
Processing

Discovered

Unseen

Queue: G → J → D → E

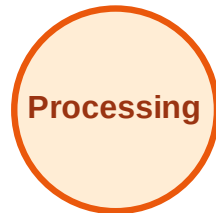
Visited: A, B, C



BFS Traversal

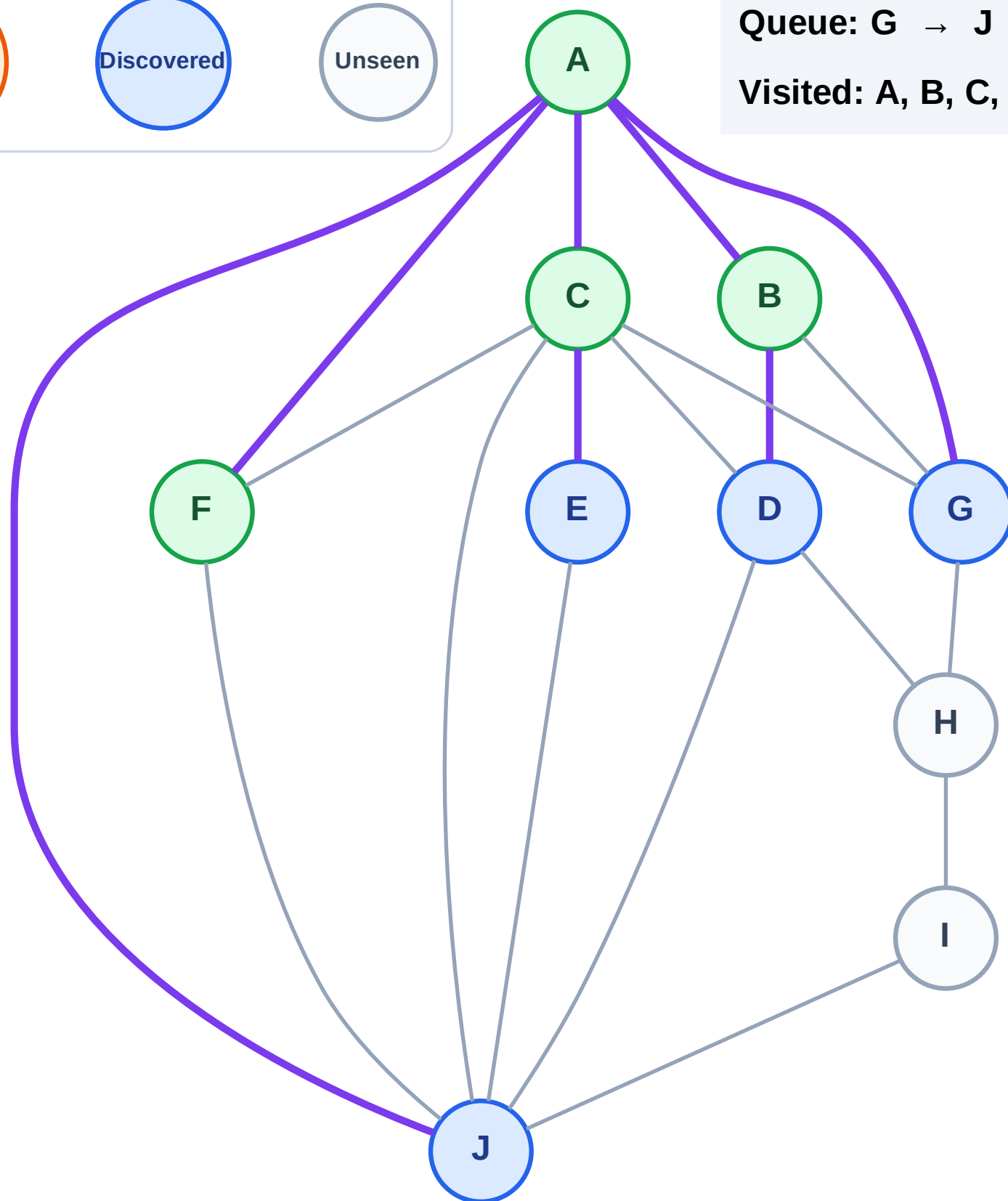
Step 9: No new neighbors from F

Legend



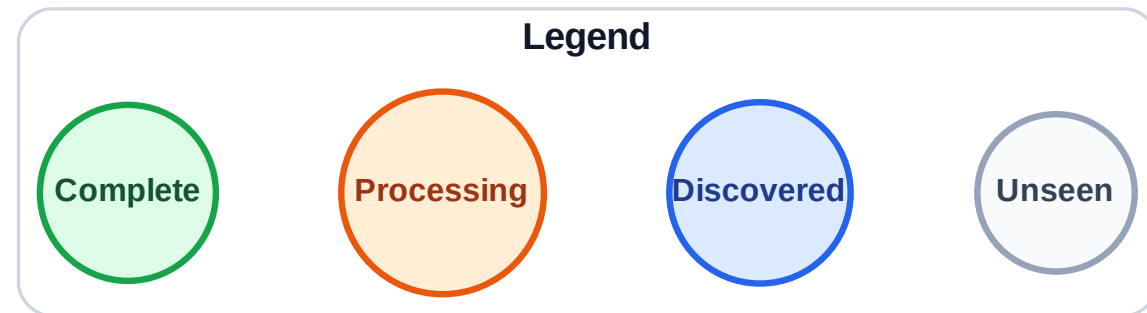
Queue: G → J → D → E

Visited: A, B, C, F

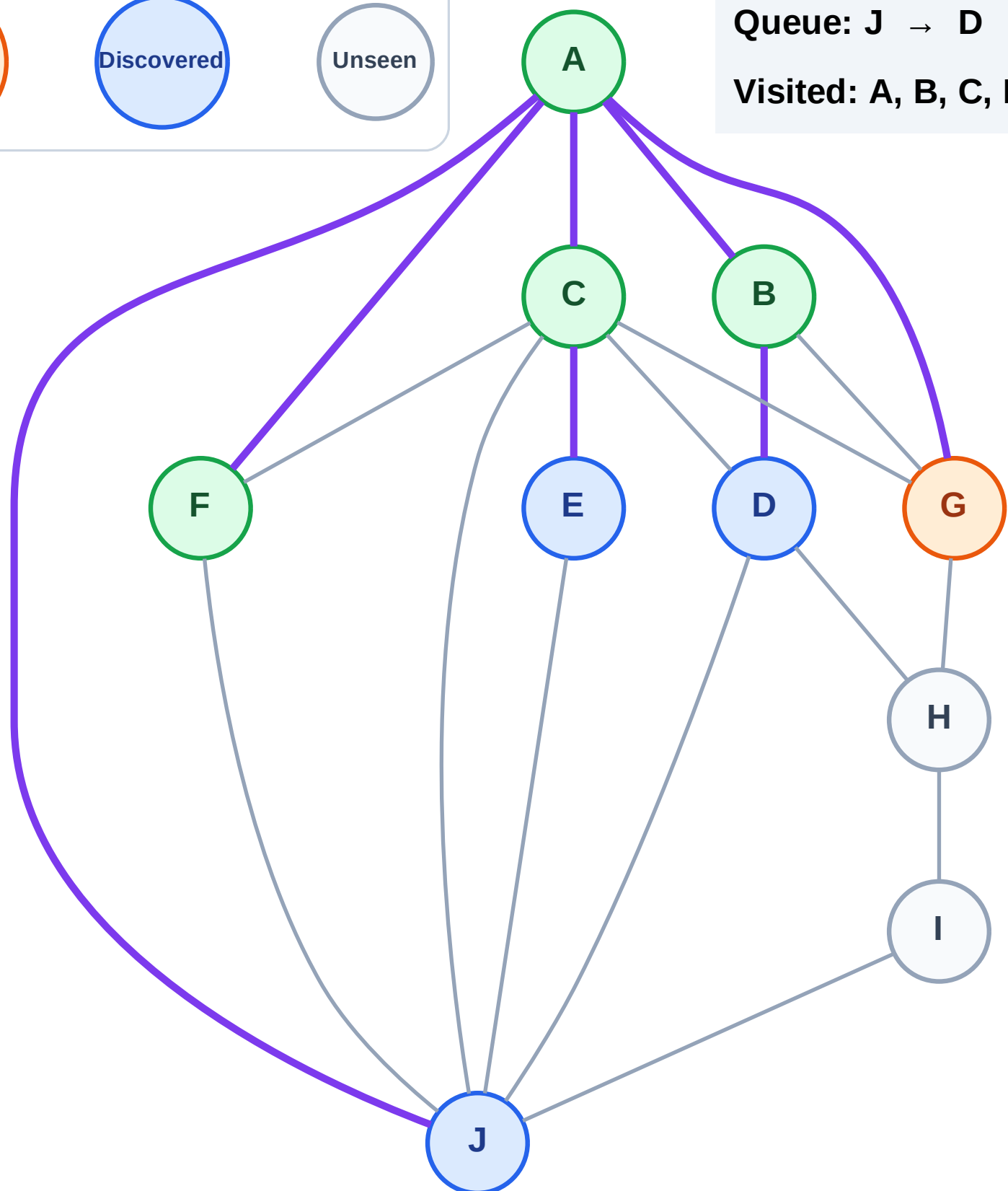


BFS Traversal

Step 10: Process node G



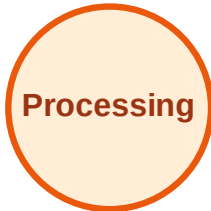
Queue: J → D → E
Visited: A, B, C, F



BFS Traversal

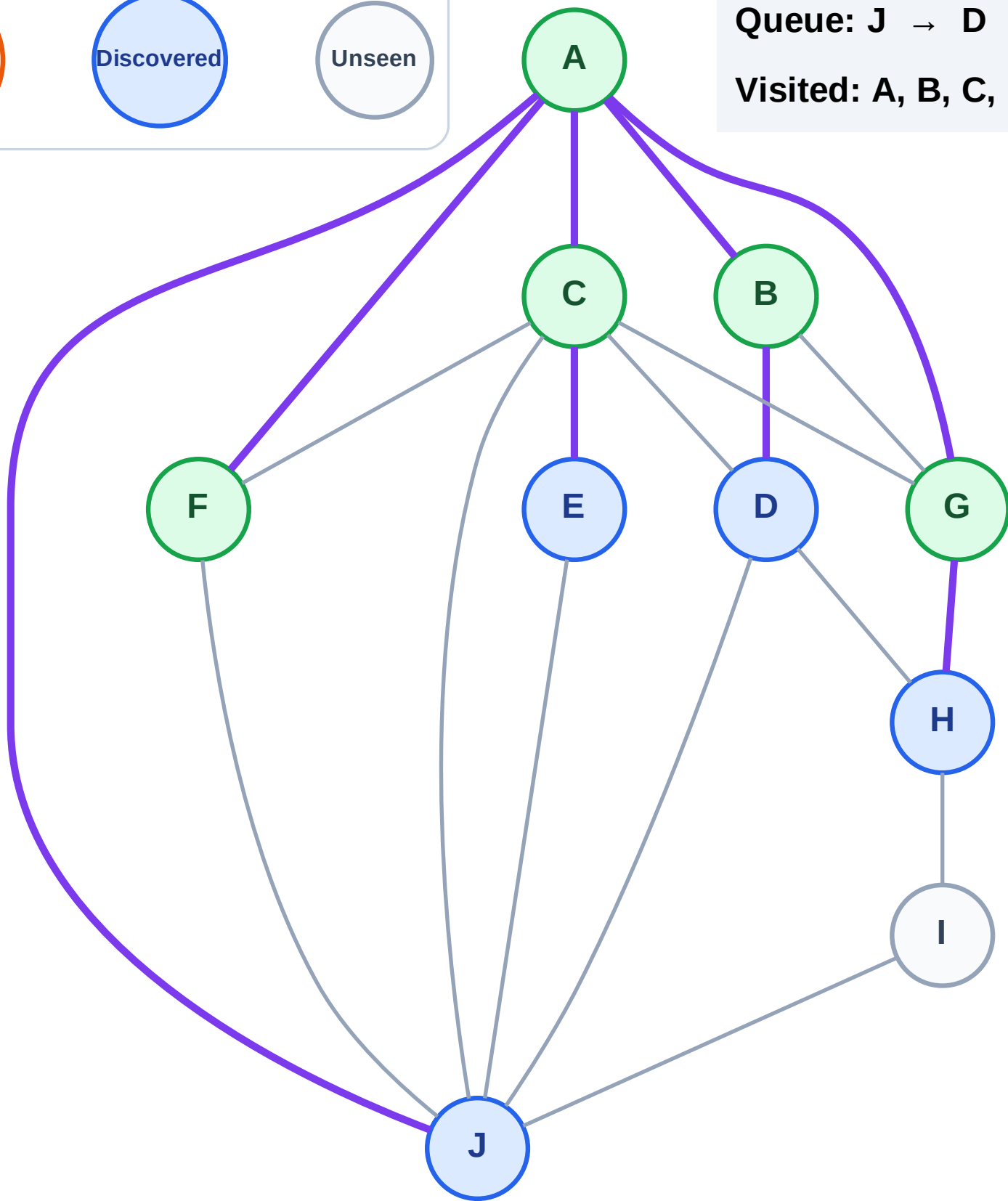
Step 11: Discover H from G

Legend



Queue: J → D → E → H

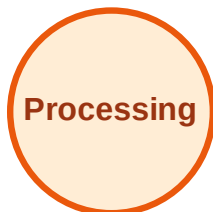
Visited: A, B, C, F, G



BFS Traversal

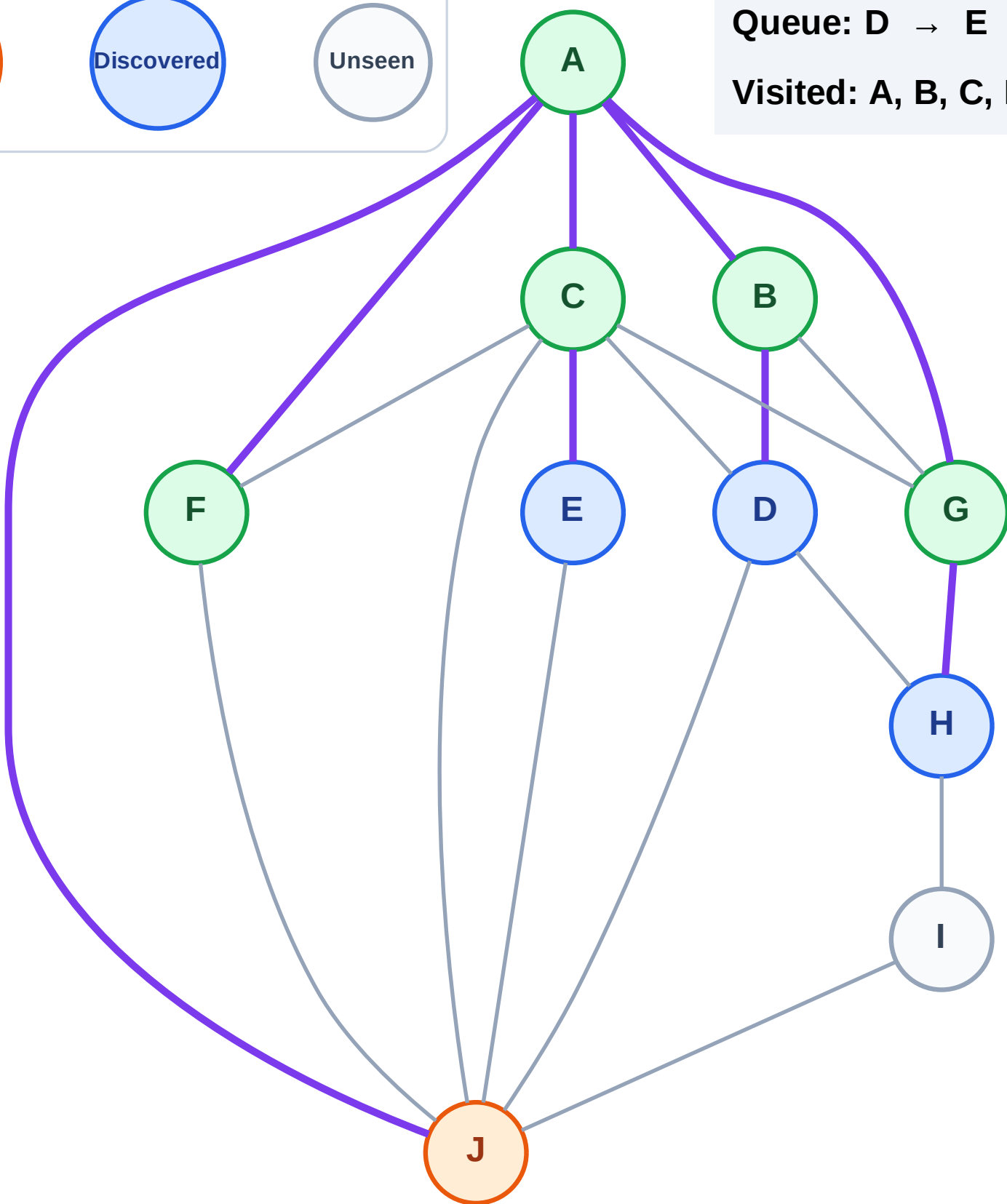
Step 12: Process node J

Legend



Queue: D → E → H

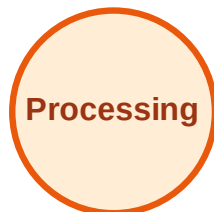
Visited: A, B, C, F, G



BFS Traversal

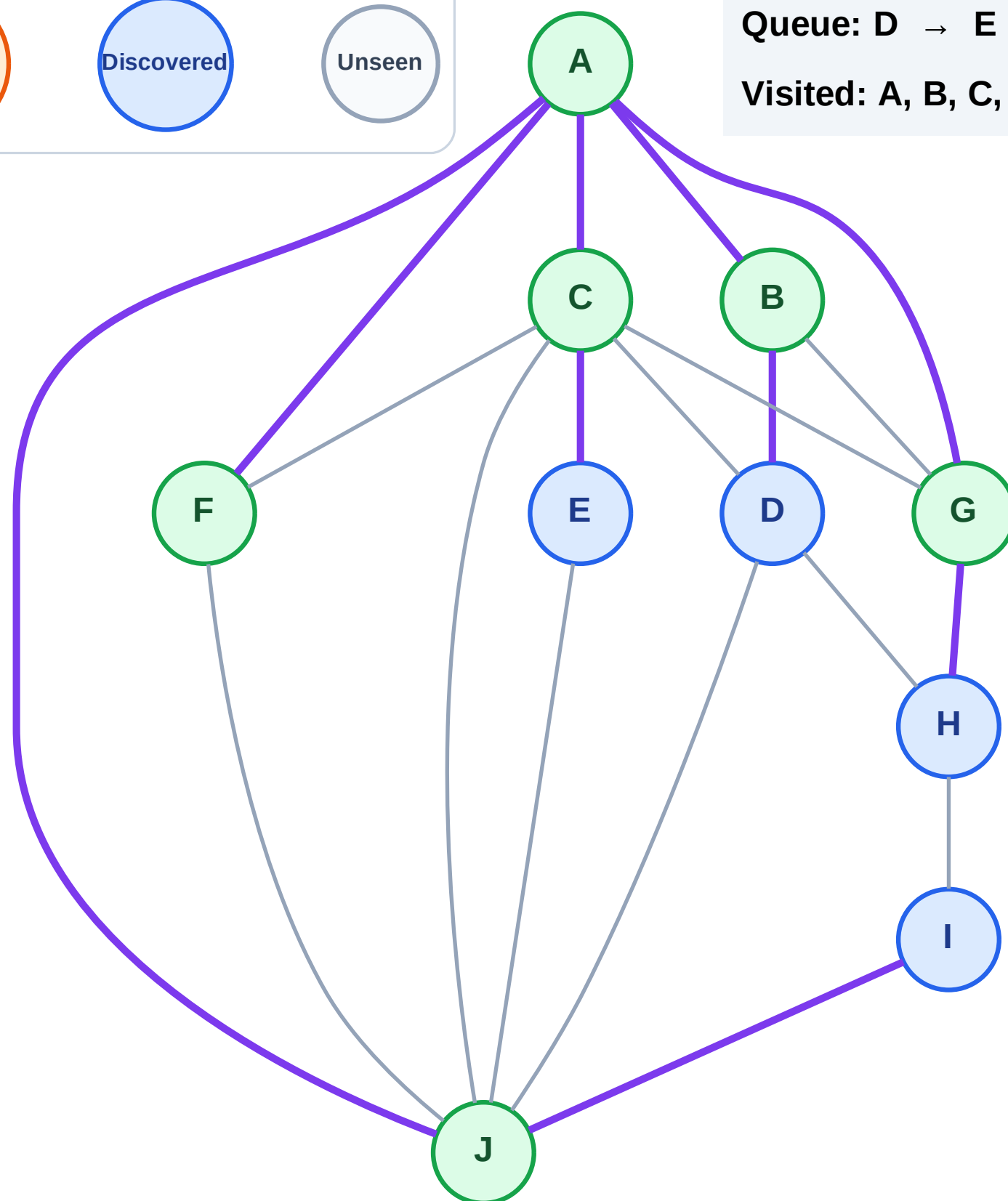
Step 13: Discover I from J

Legend



Queue: D → E → H → I

Visited: A, B, C, F, G, J



BFS Traversal

Step 14: Process node D

Legend

Complete

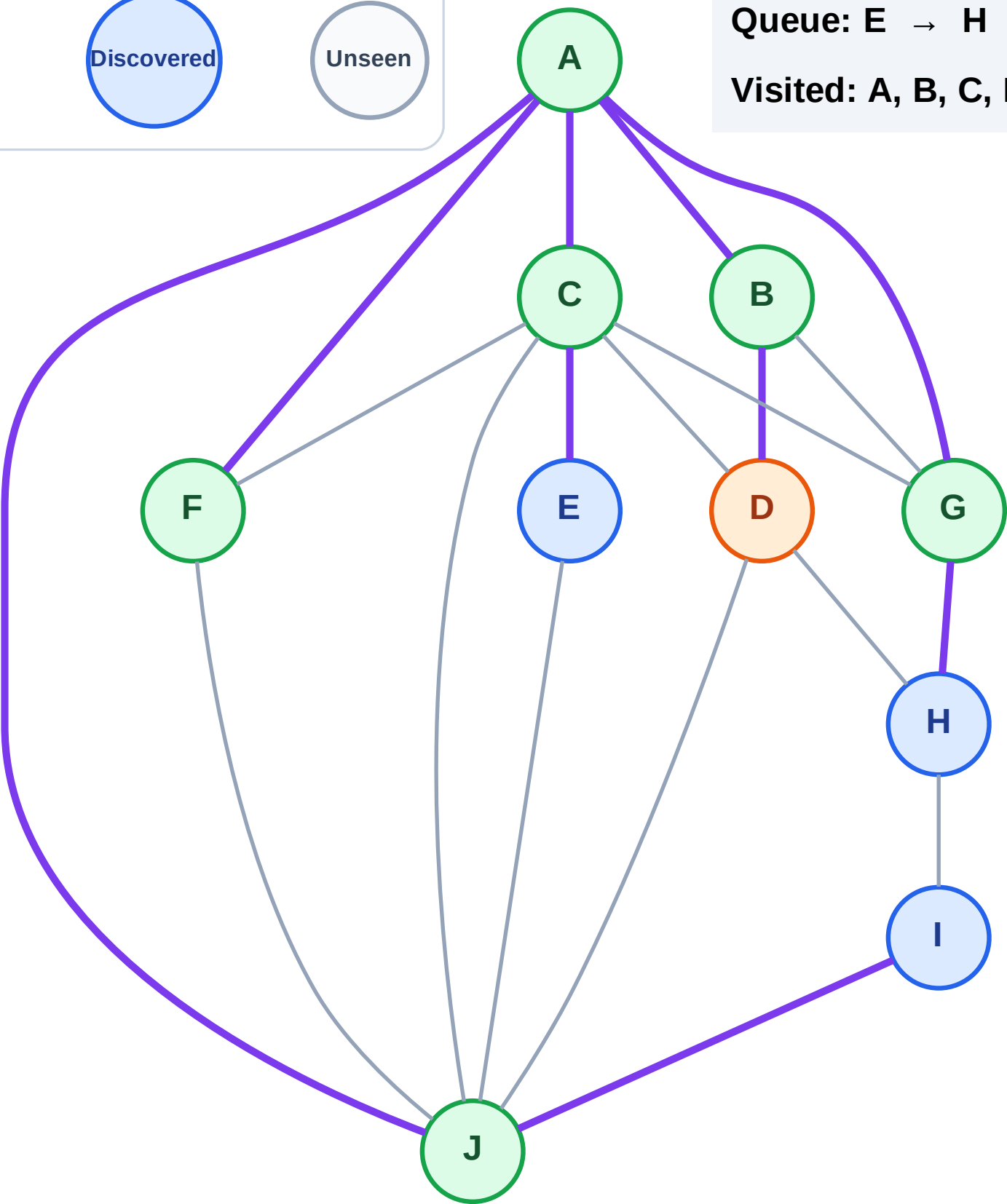
Processing

Discovered

Unseen

Queue: E → H → I

Visited: A, B, C, F, G, J



BFS Traversal

Step 15: No new neighbors from D

Legend

Complete

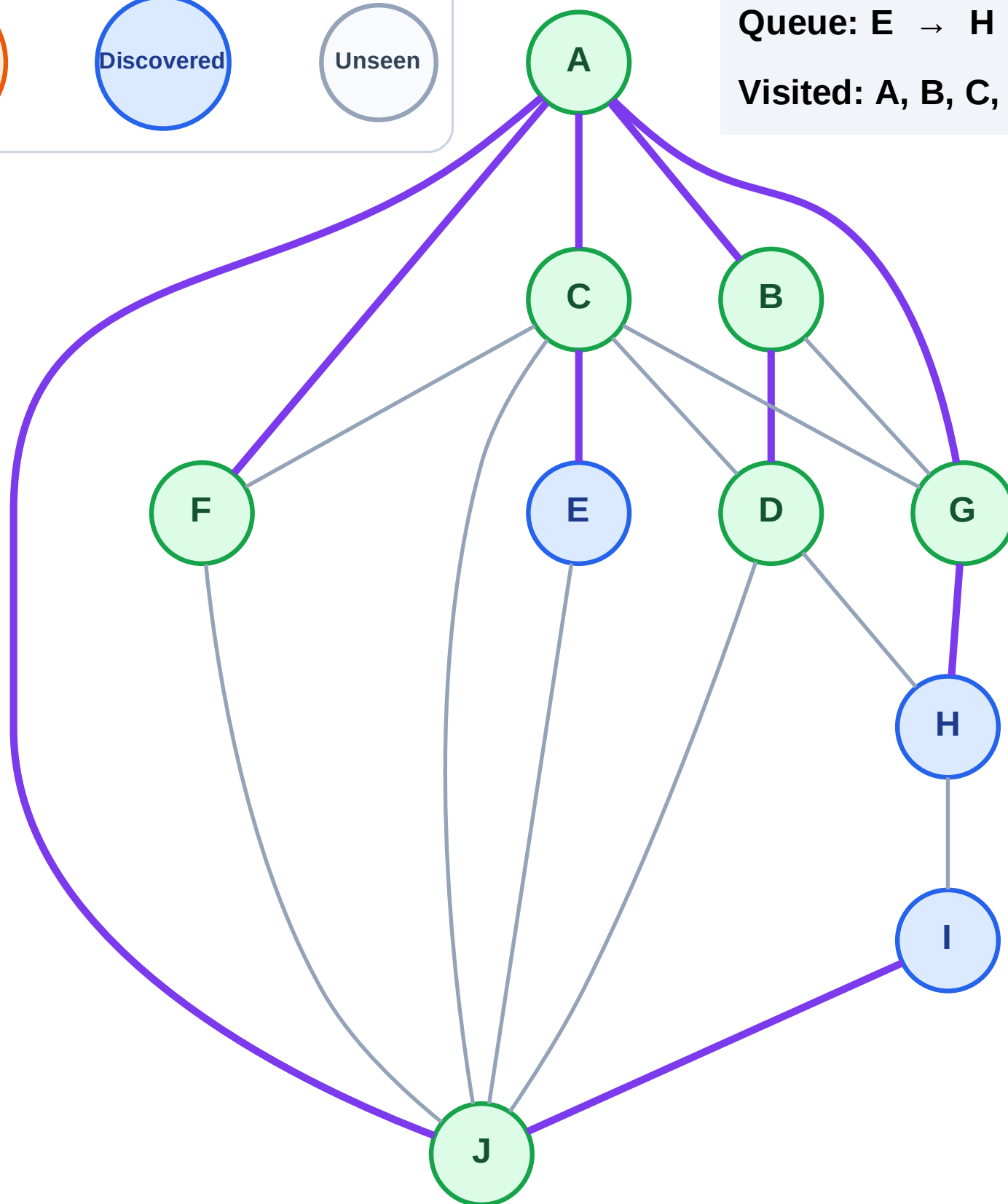
Processing

Discovered

Unseen

Queue: E → H → I

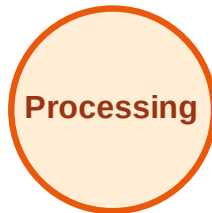
Visited: A, B, C, D, F, G, J



BFS Traversal

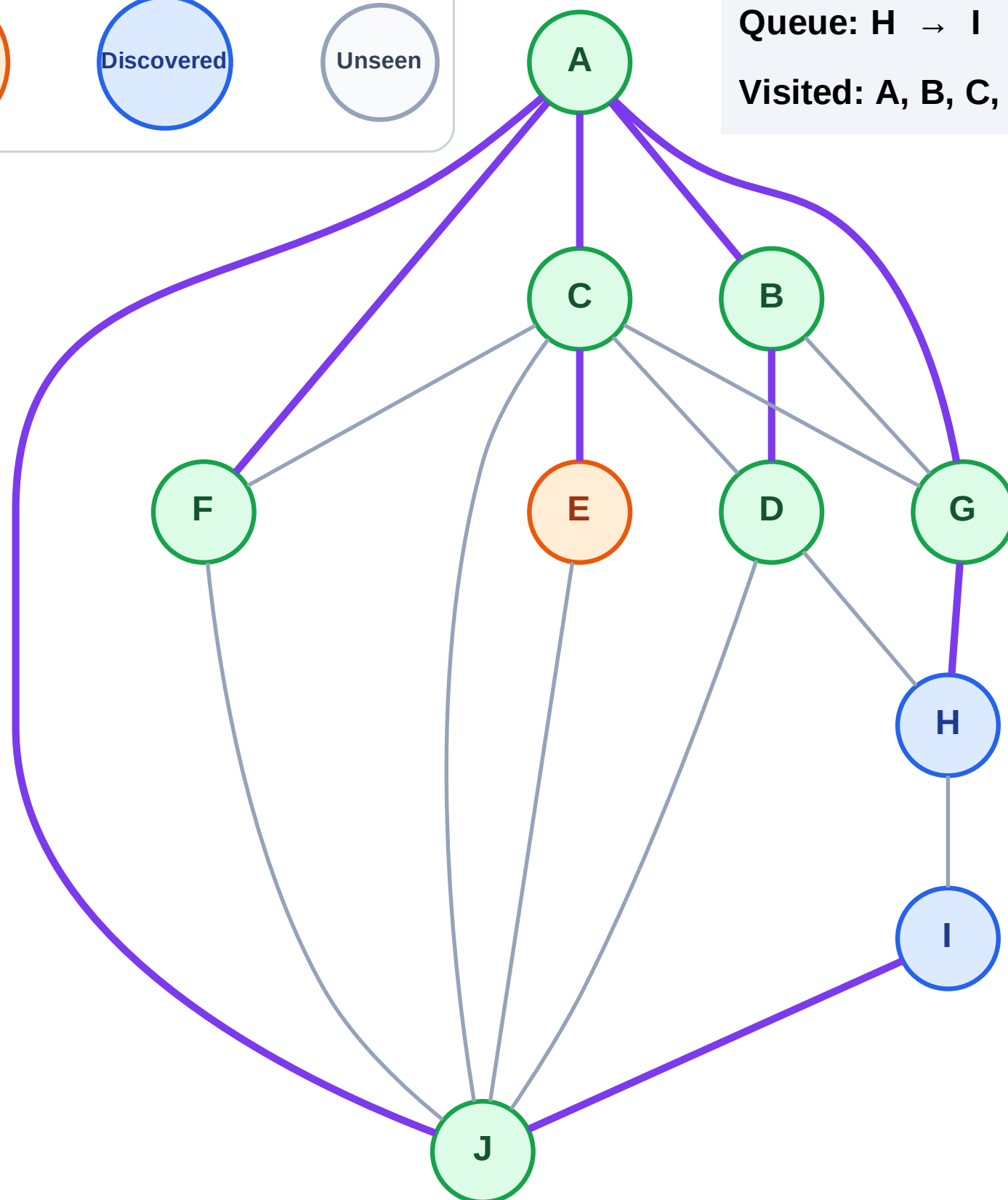
Step 16: Process node E

Legend



Queue: H → I

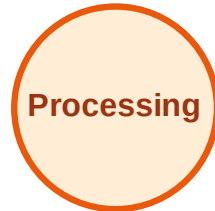
Visited: A, B, C, D, F, G, J



BFS Traversal

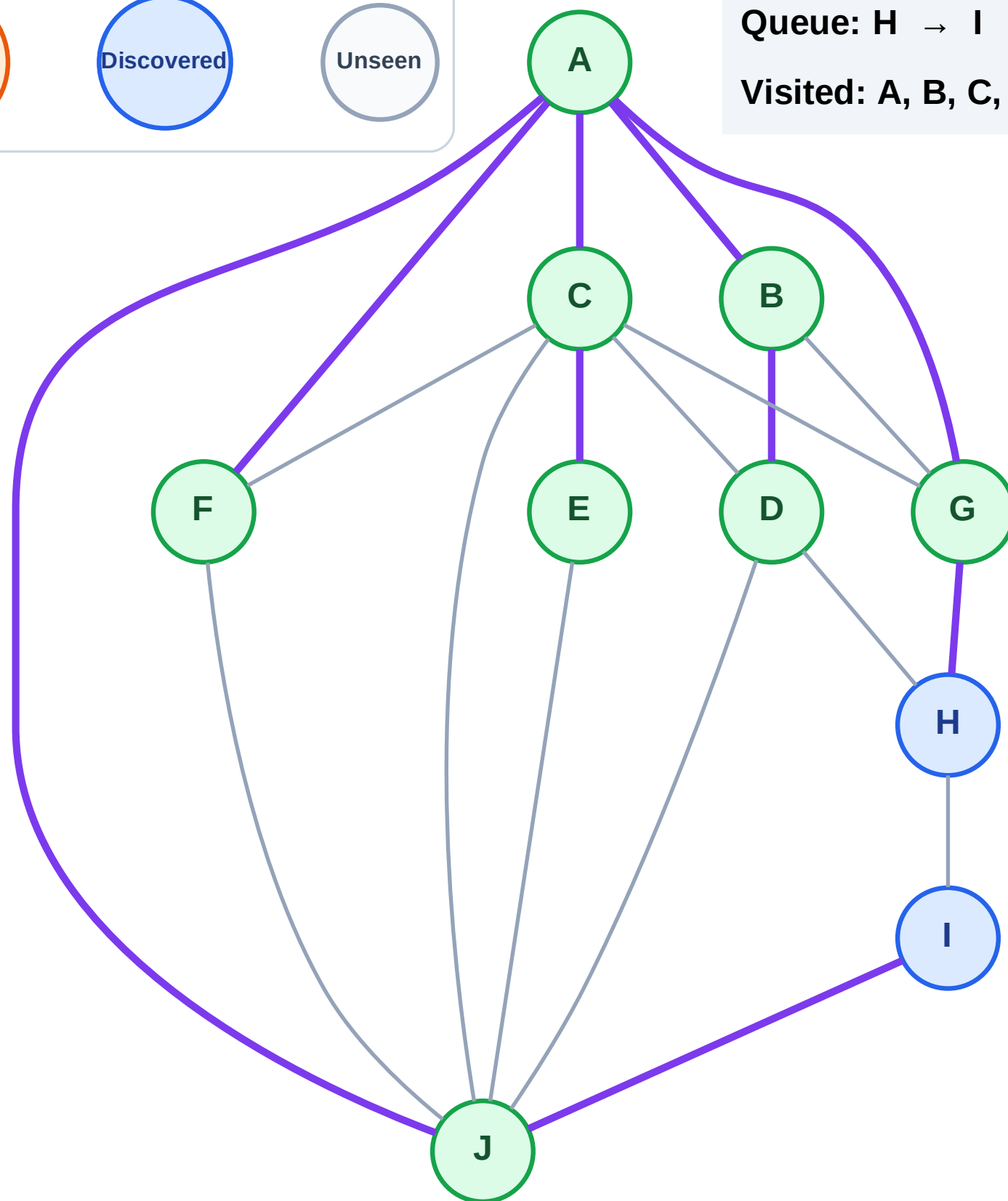
Step 17: No new neighbors from E

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

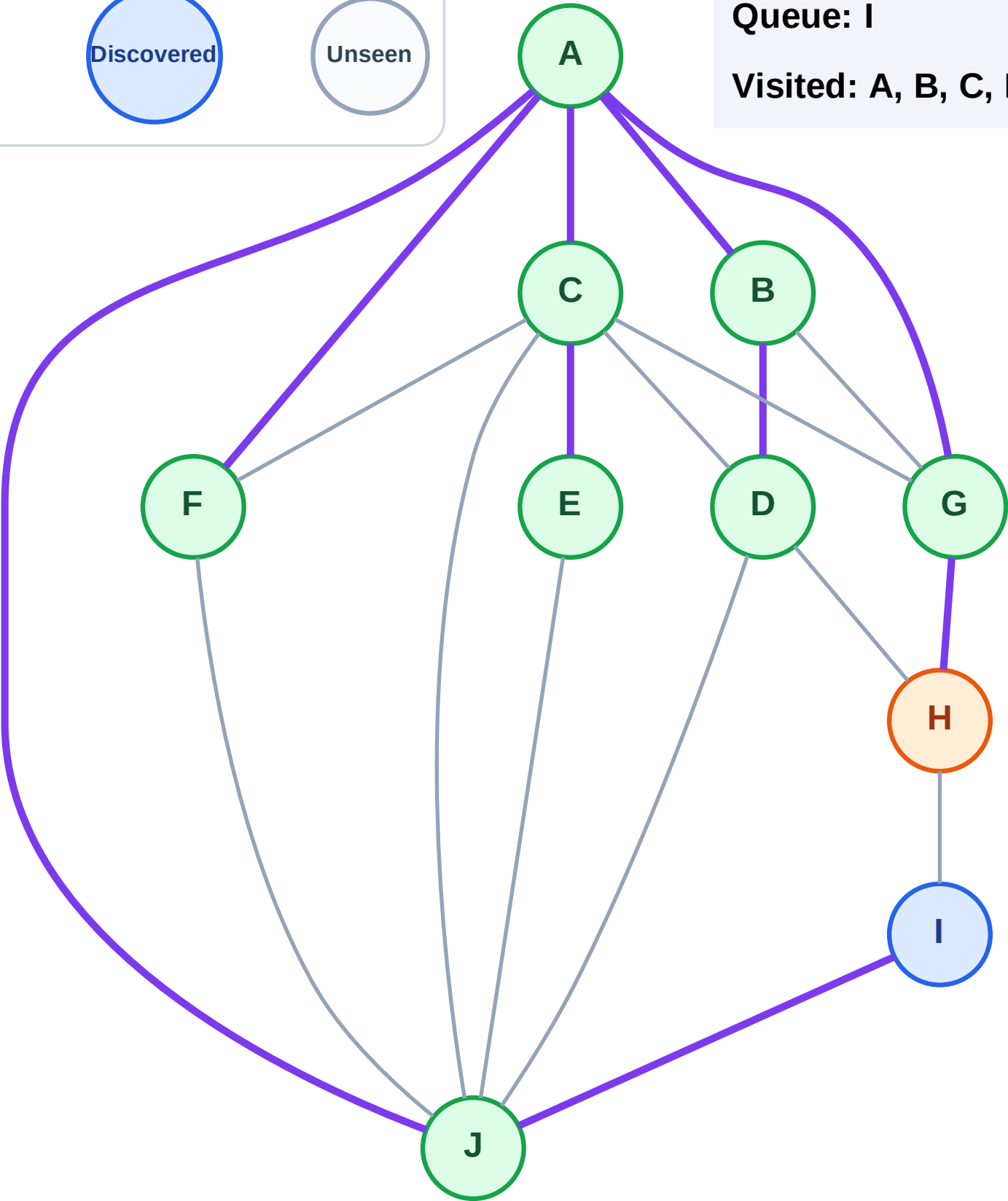
Processing

Discovered

Unseen

Queue: I

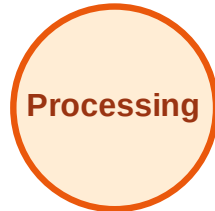
Visited: A, B, C, D, E, F, G, J



BFS Traversal

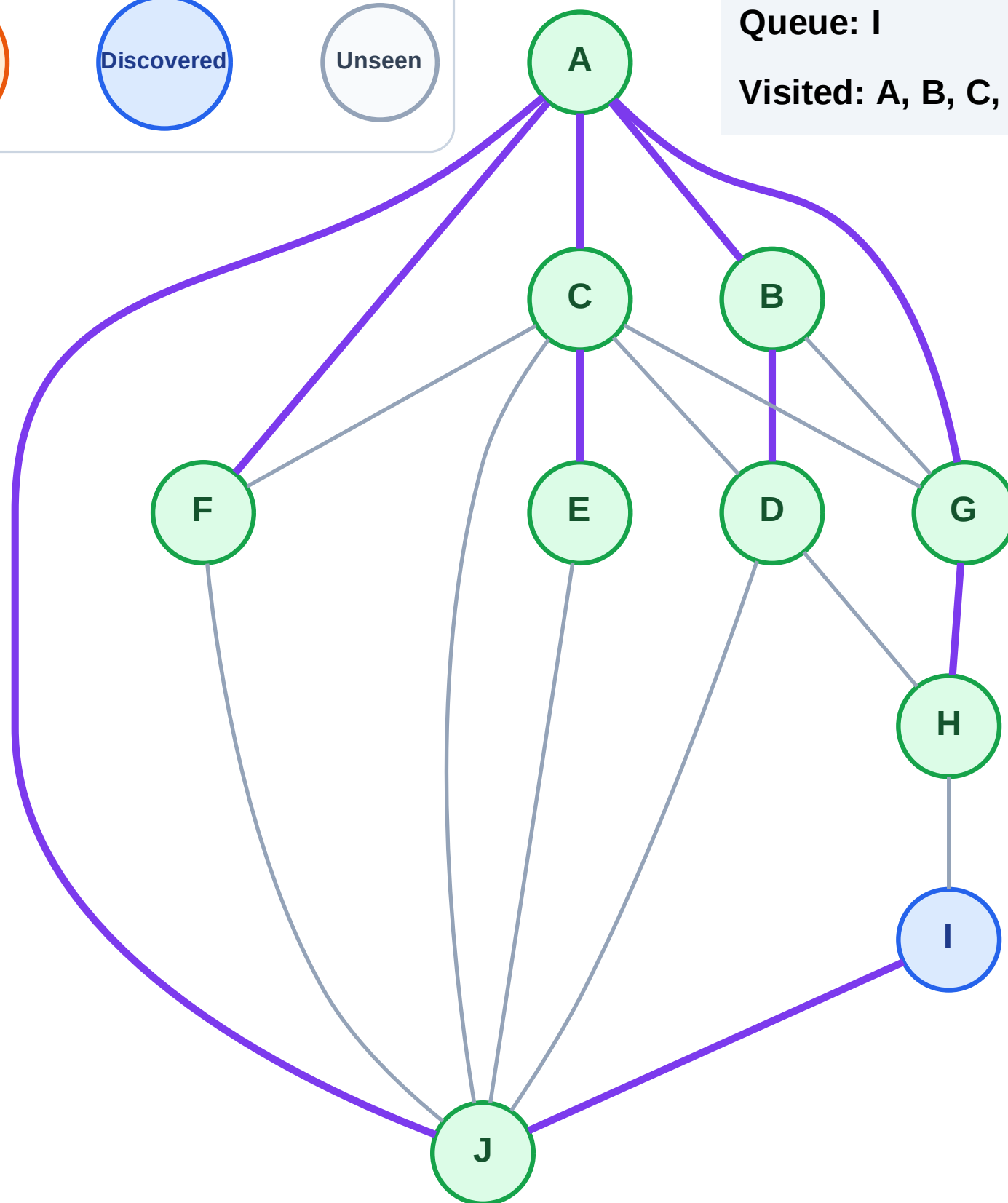
Step 19: No new neighbors from H

Legend



Queue: I

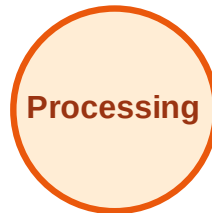
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

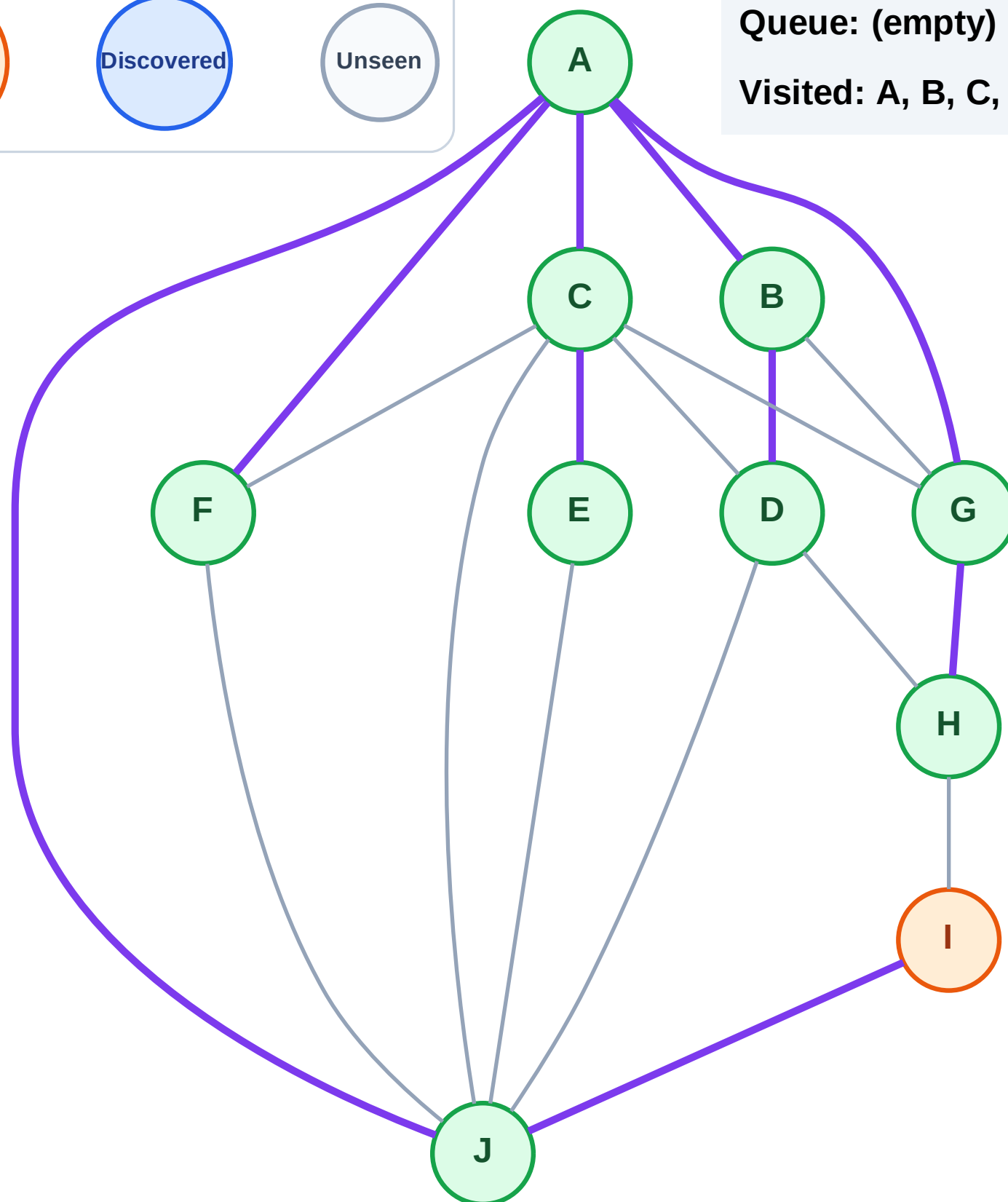
Step 20: Process node I

Legend



Queue: (empty)

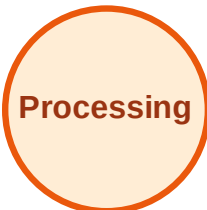
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

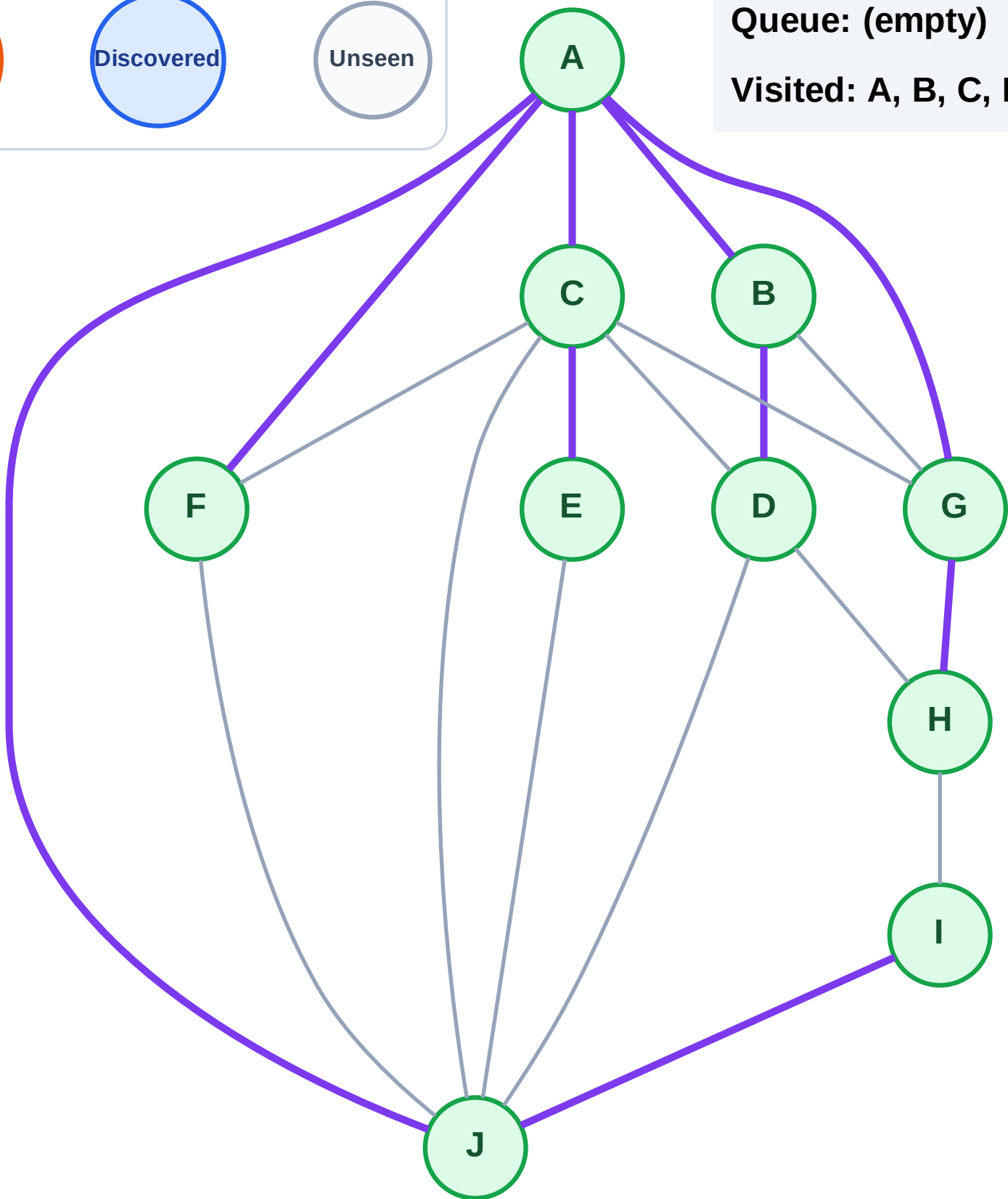
Step 21: No new neighbors from I

Legend



Queue: (empty)

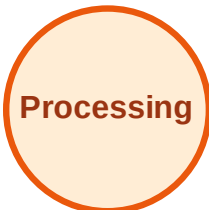
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

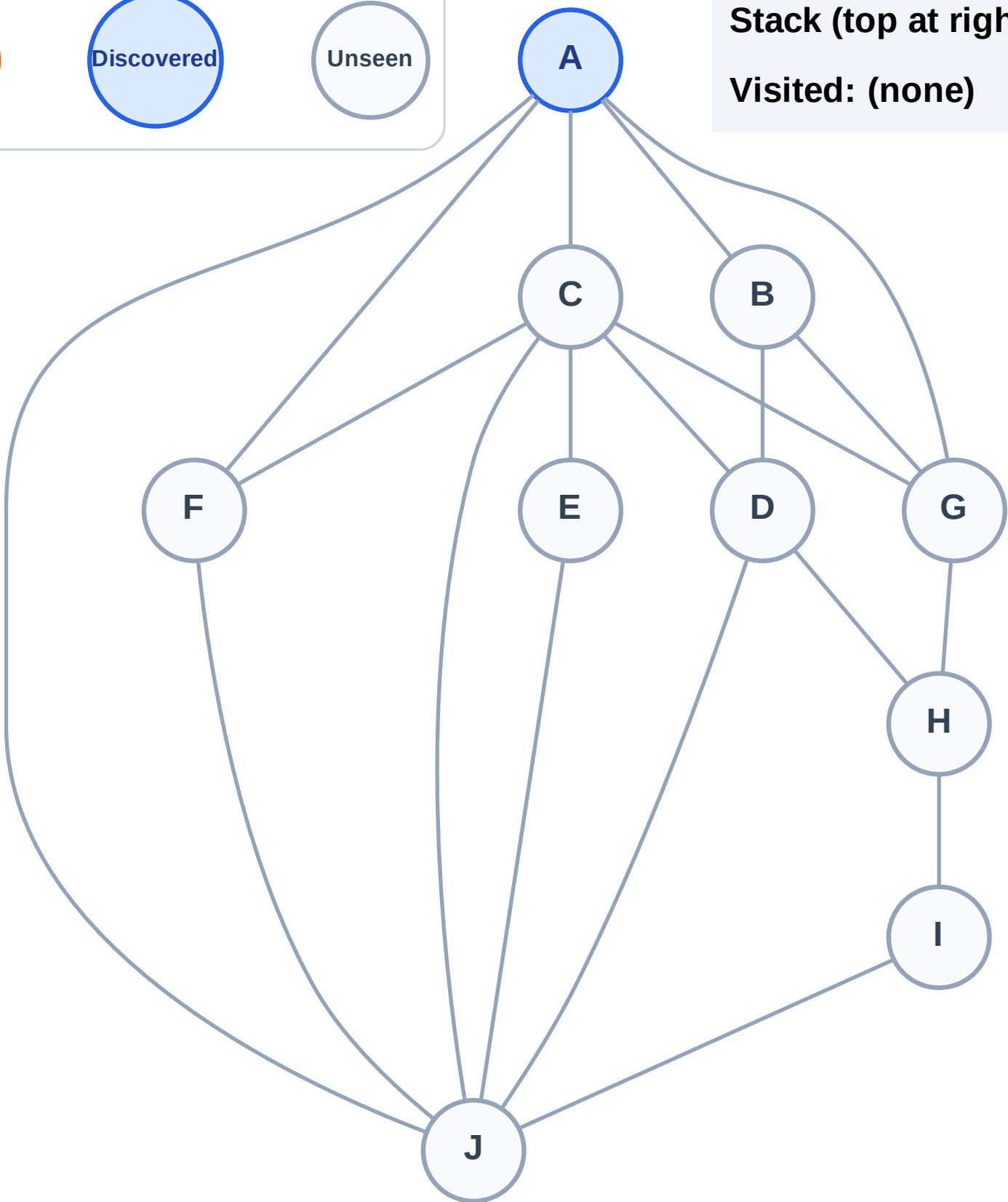
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

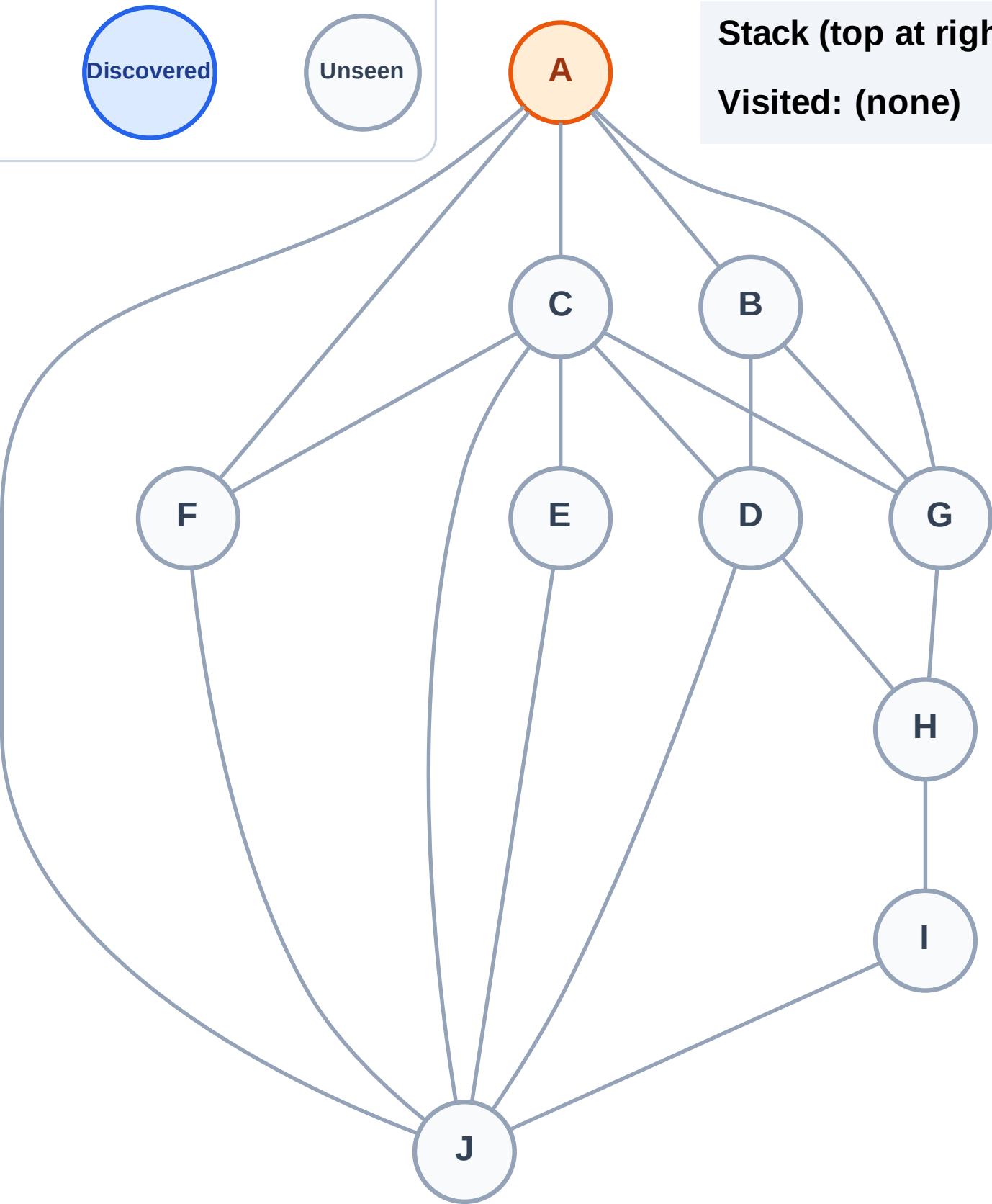
Complete

Processing

Discovered

Unseen

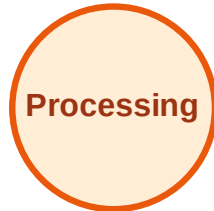
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

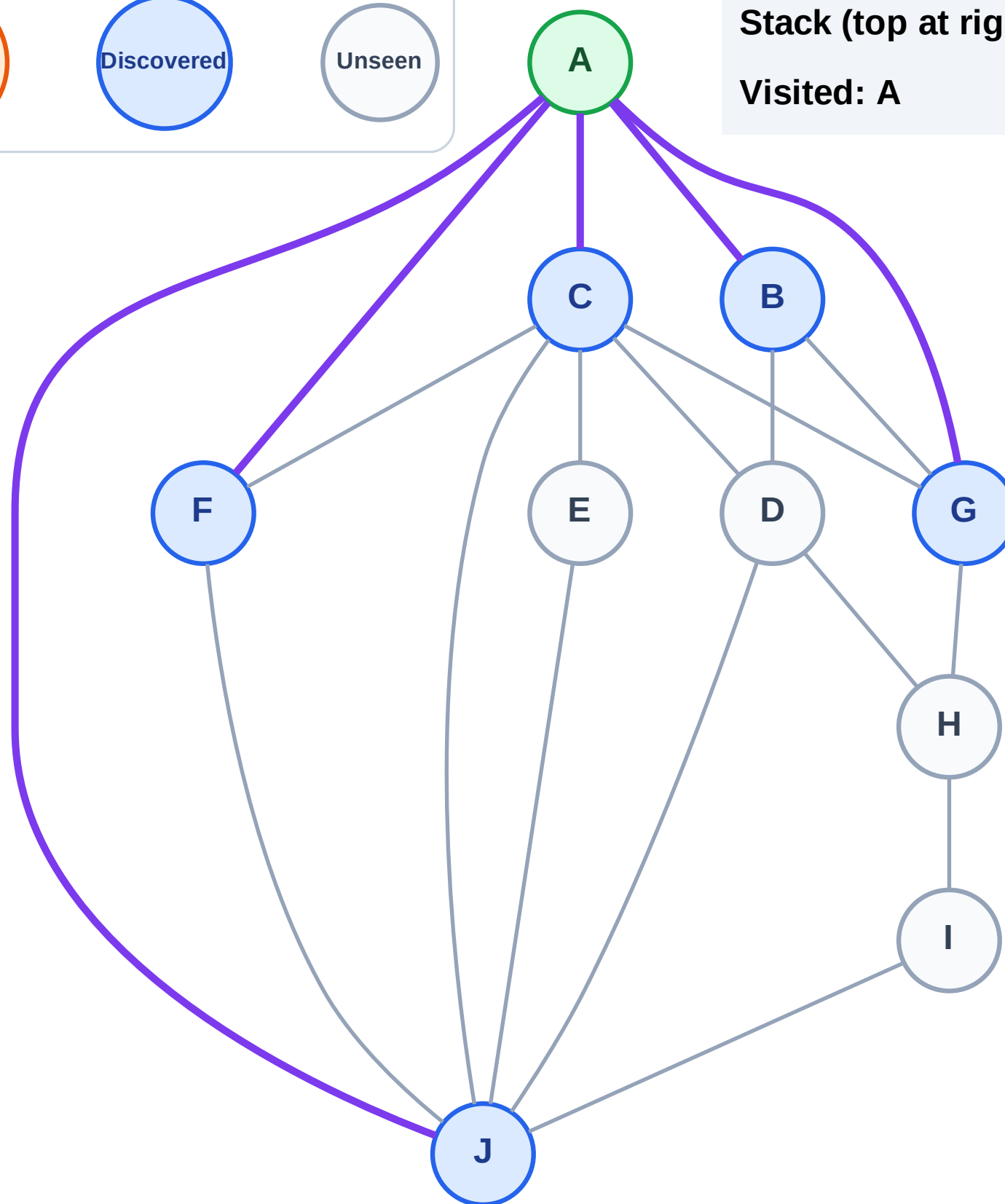
Step 3: Discover J, G, F, C, B from A

Legend



Stack (top at right): J | G | F | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

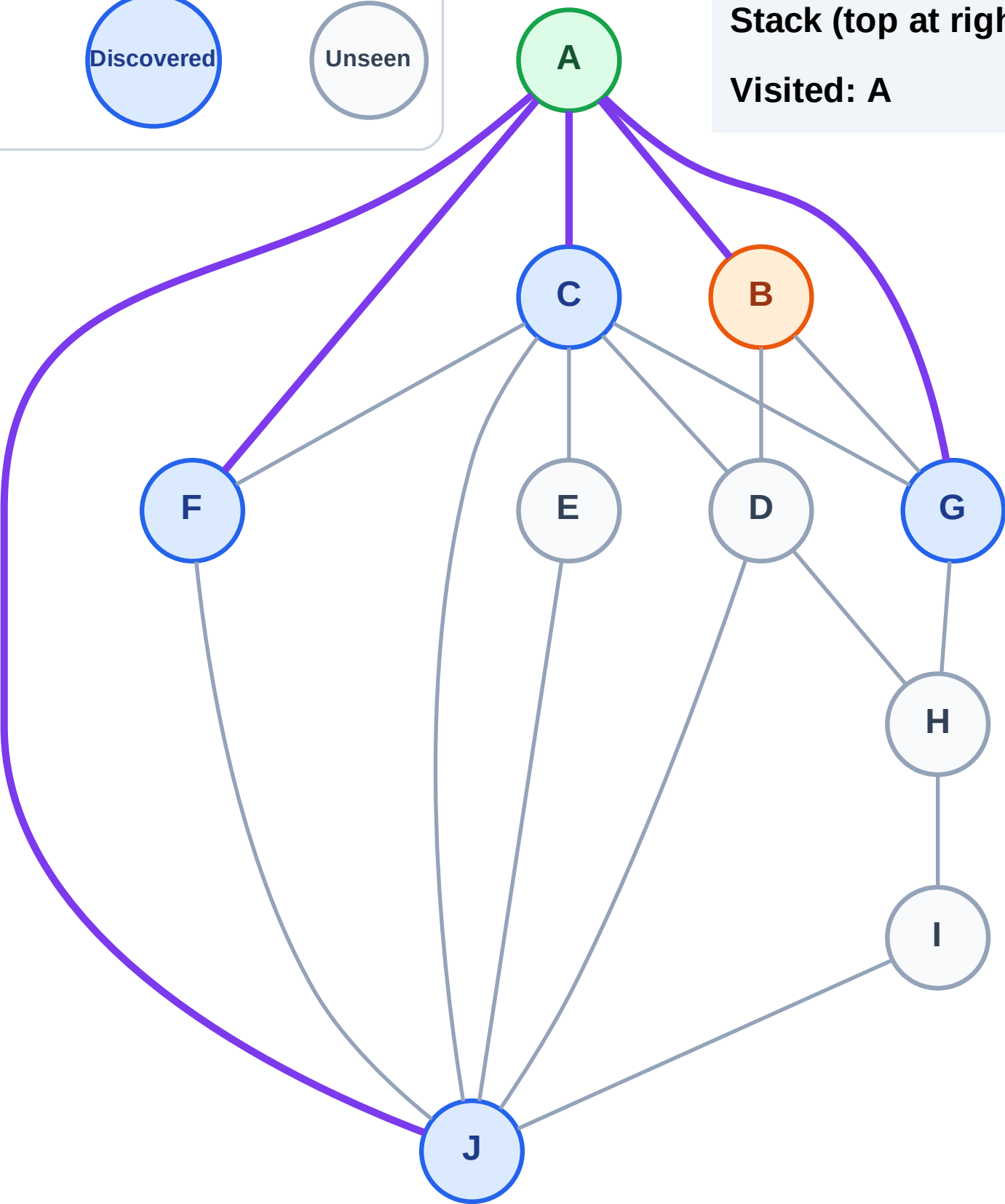
Complete

Processing

Discovered

Unseen

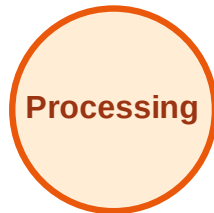
Stack (top at right): J | G | F | C
Visited: A



DFS Traversal

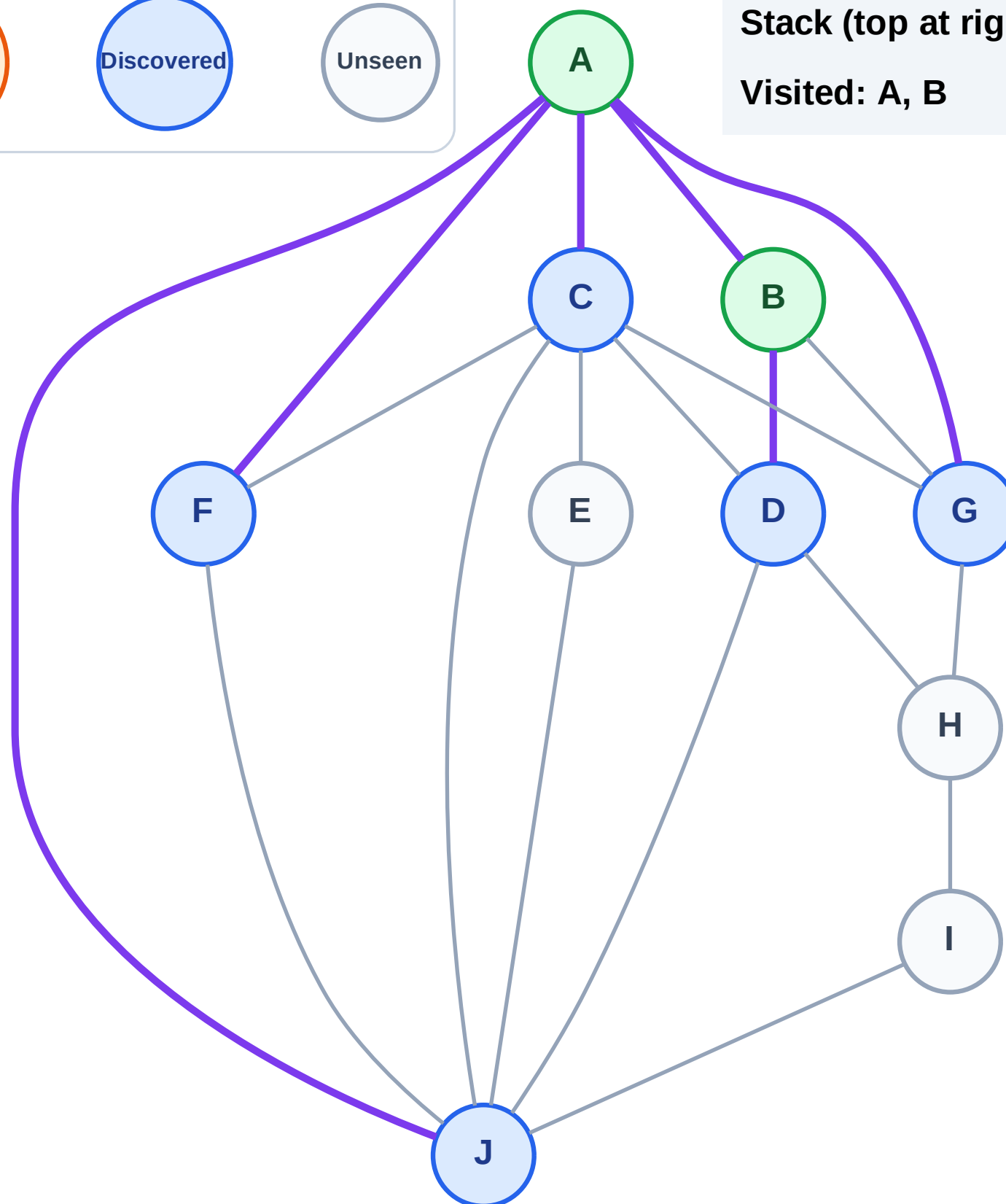
Step 5: Discover D from B

Legend



Stack (top at right): J | G | F | C | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

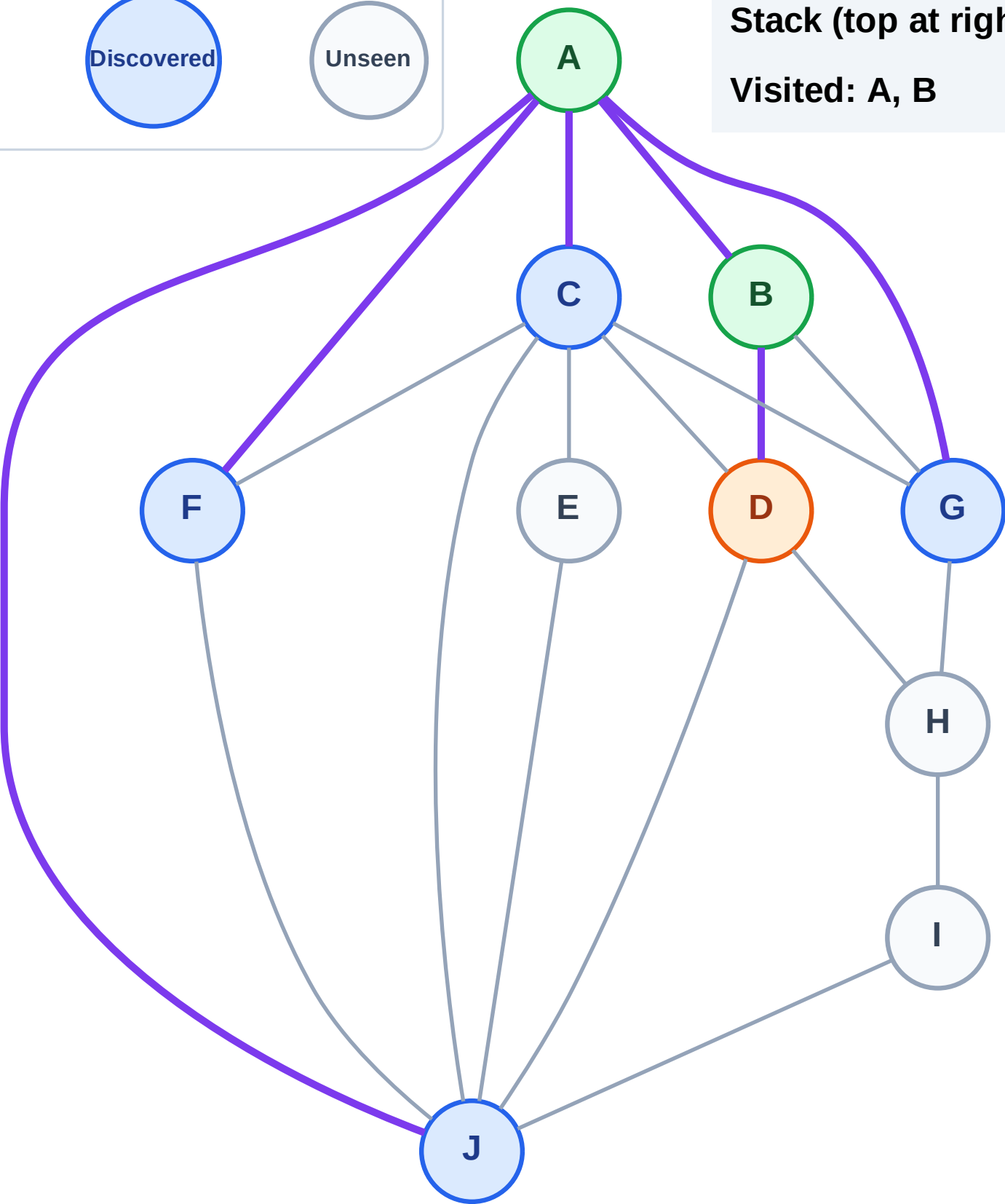
Complete

Processing

Discovered

Unseen

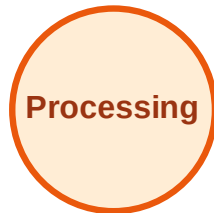
Stack (top at right): J | G | F | C
Visited: A, B



DFS Traversal

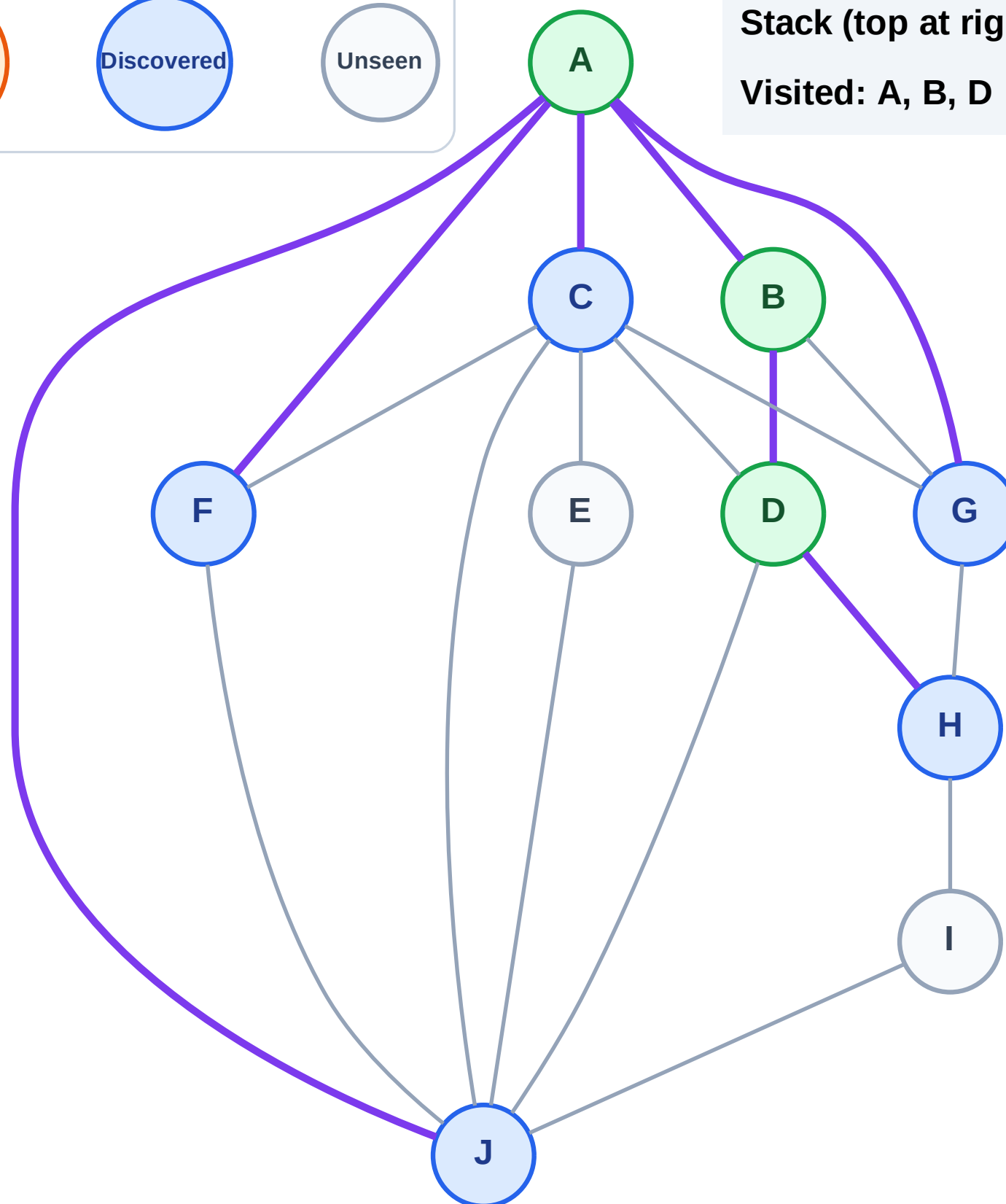
Step 7: Discover H from D

Legend



Stack (top at right): J | G | F | C | H

Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

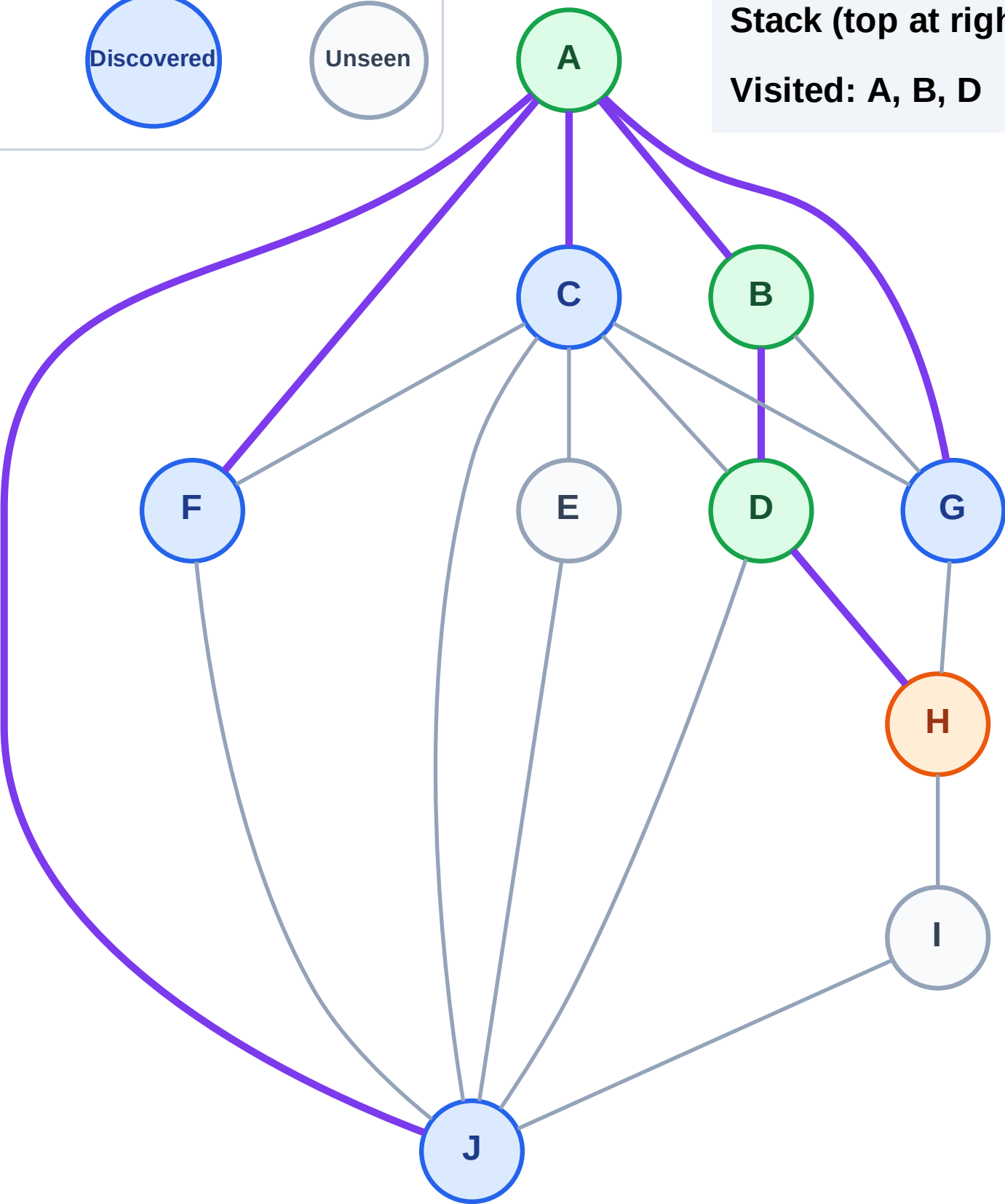
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | F | C
Visited: A, B, D



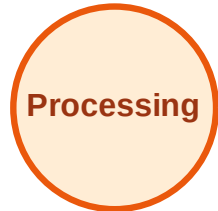
DFS Traversal

Step 9: Discover I from H

Legend



Complete



Processing



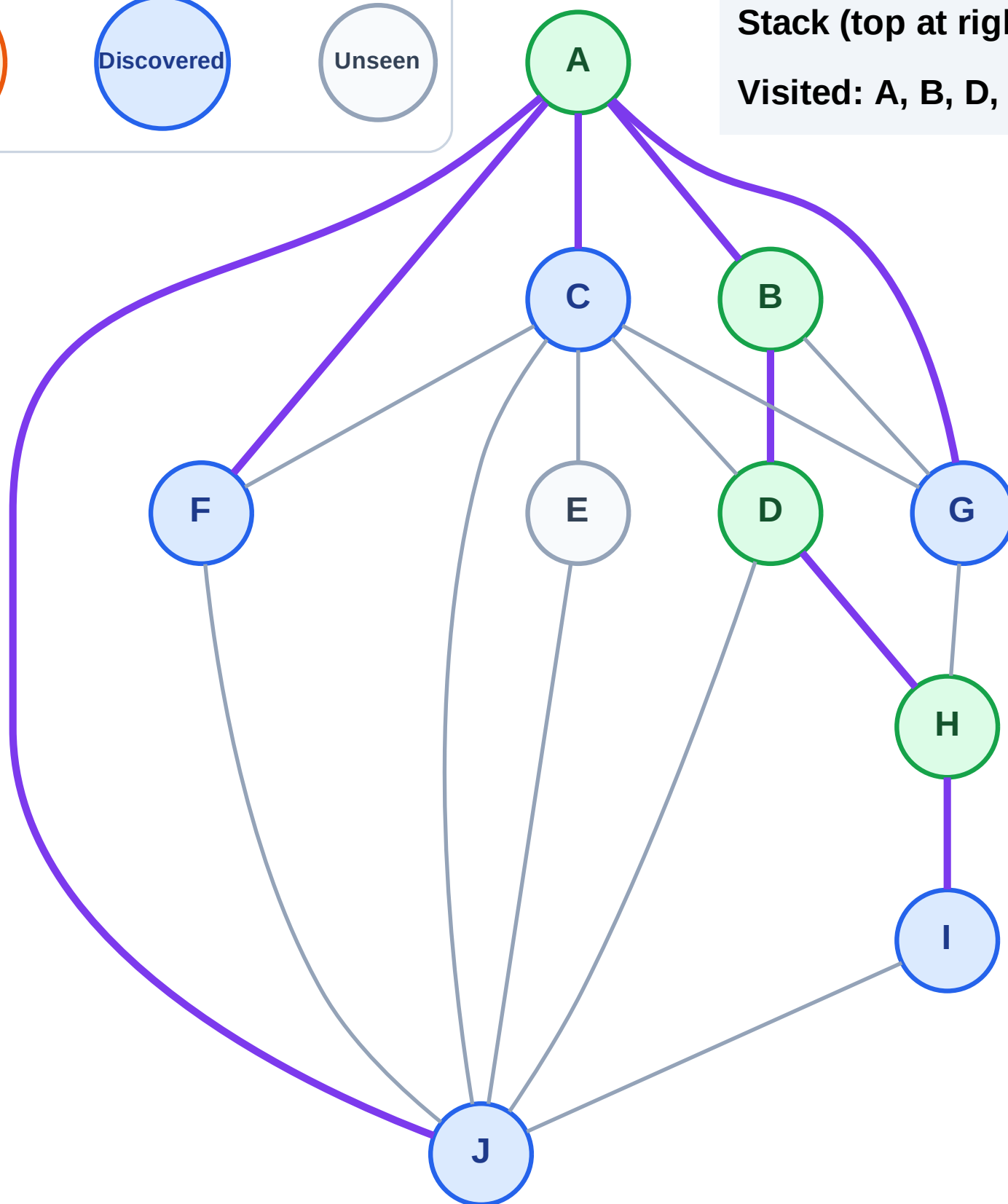
Discovered



Unseen

Stack (top at right): J | G | F | C | I

Visited: A, B, D, H



DFS Traversal

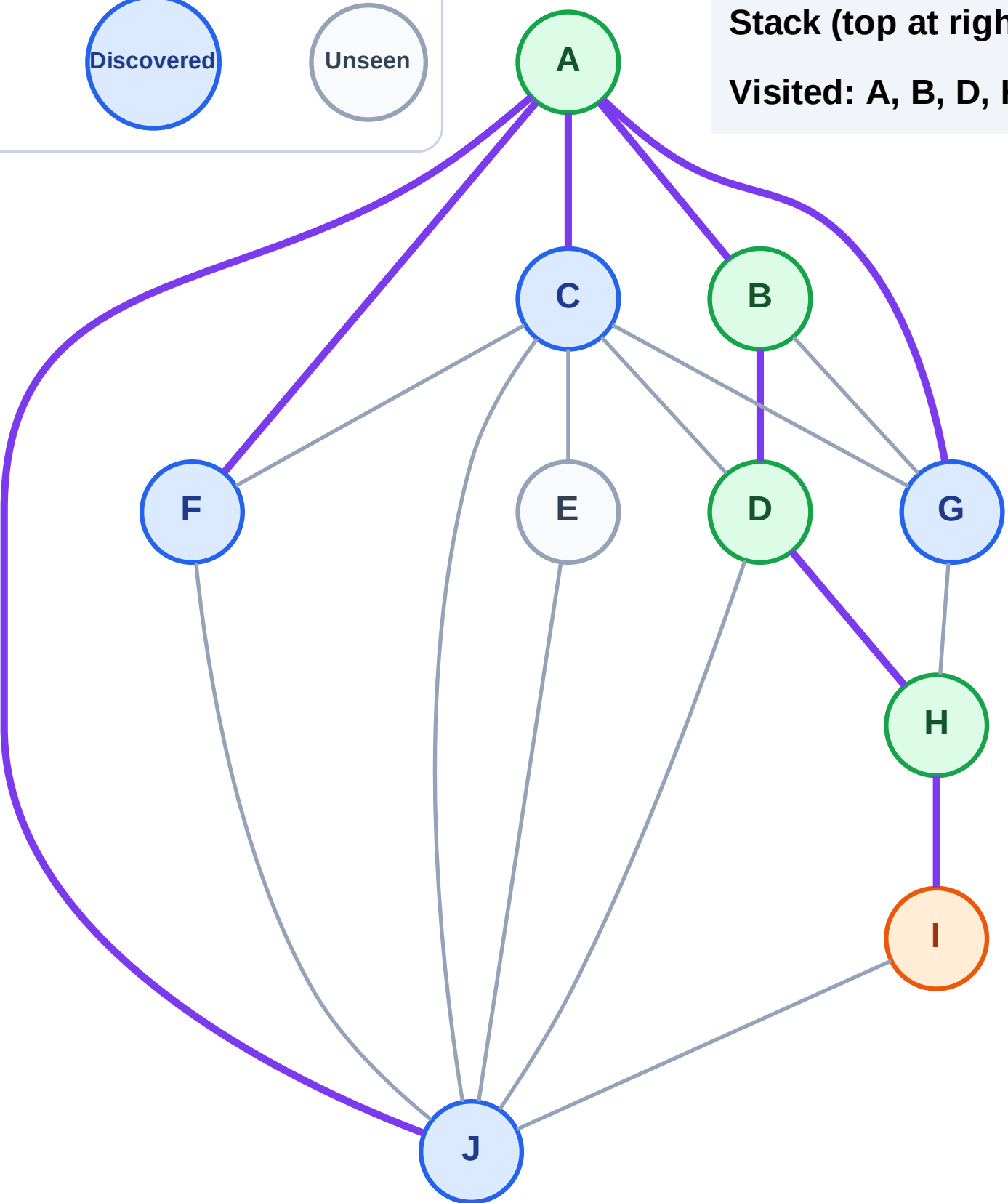
Step 10: Process node I

Legend



Stack (top at right): J | G | F | C

Visited: A, B, D, H



DFS Traversal

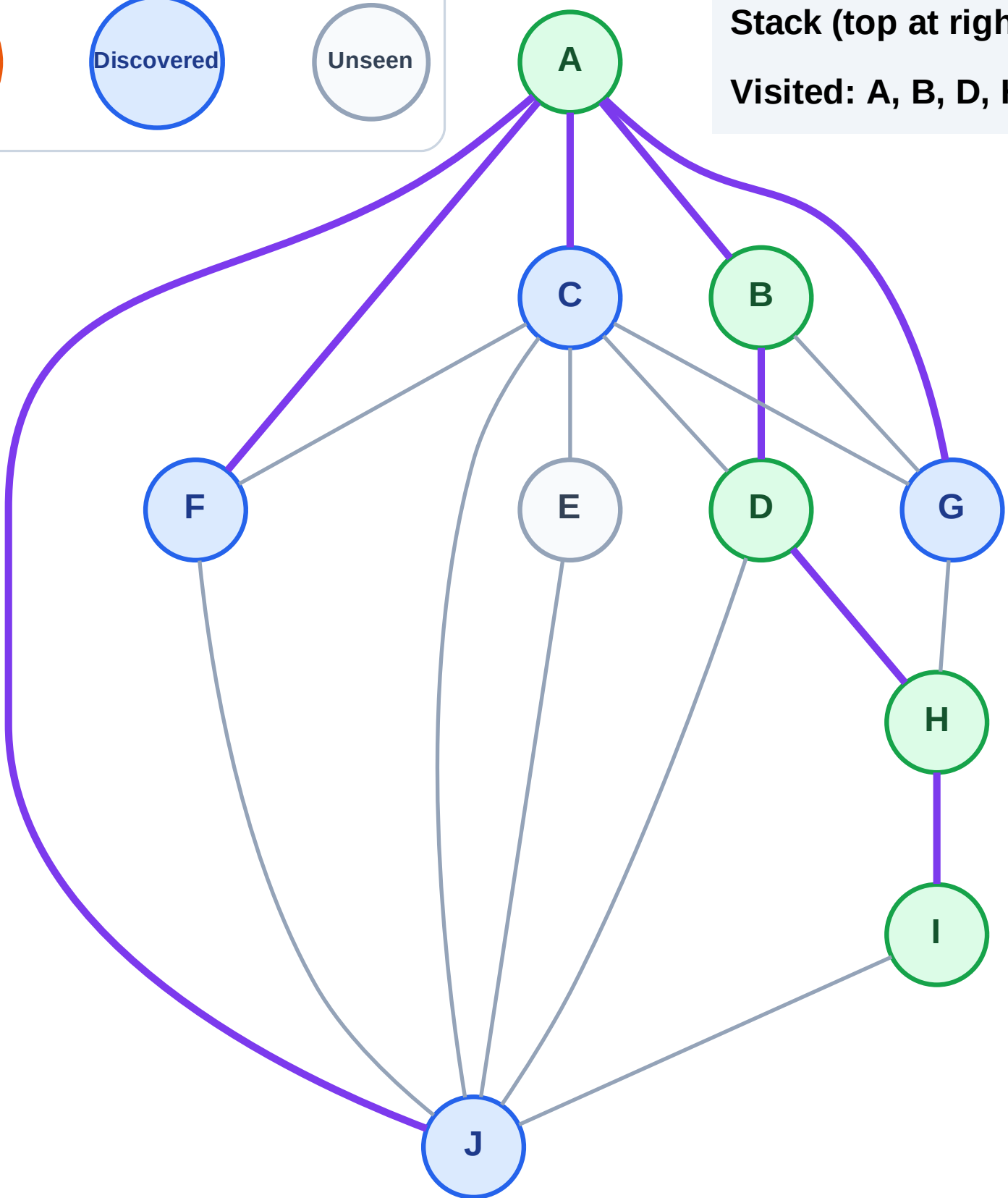
Step 11: No new neighbors from I

Legend



Stack (top at right): J | G | F | C

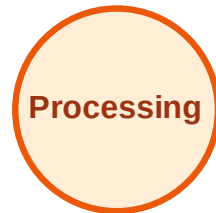
Visited: A, B, D, H, I



DFS Traversal

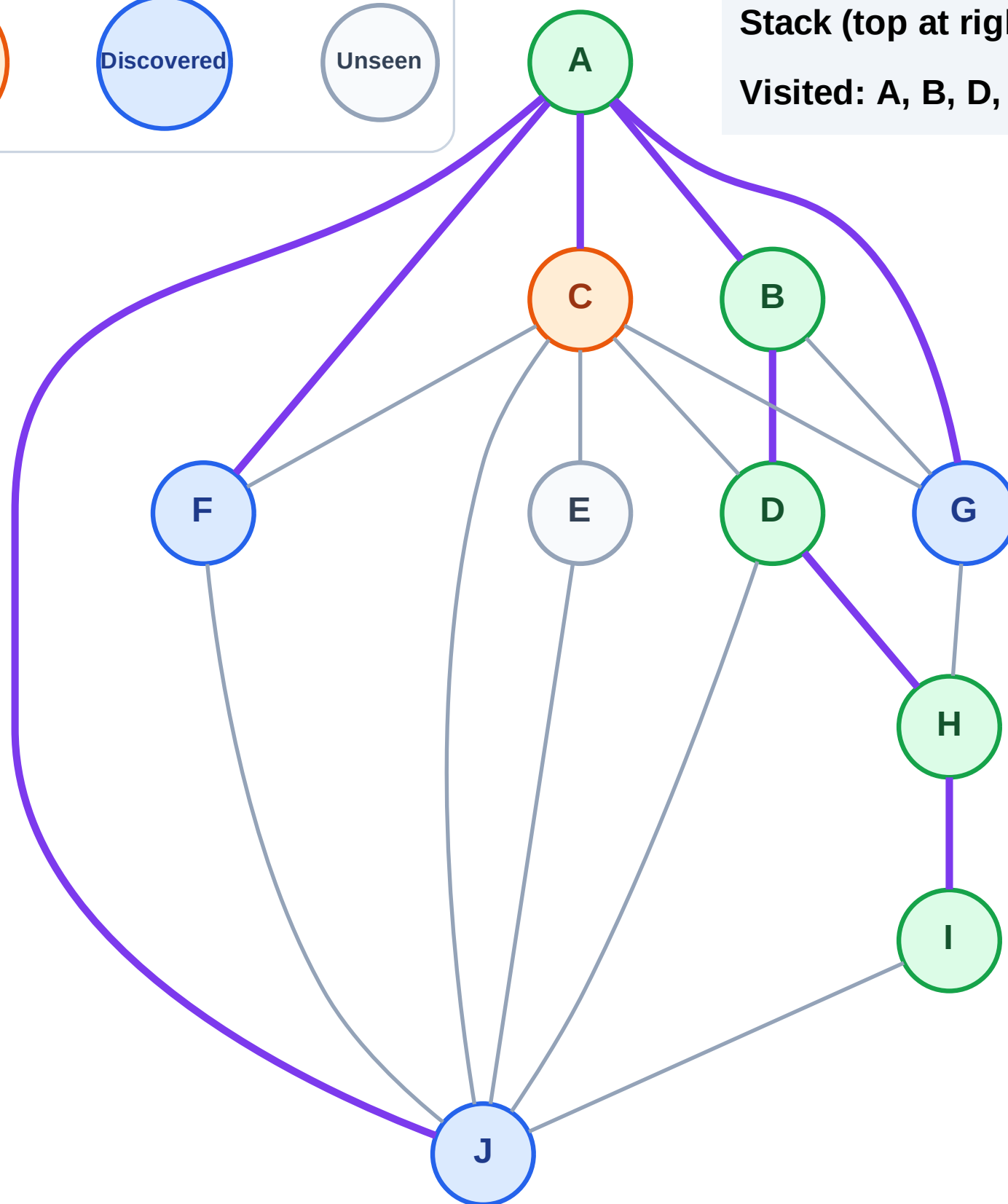
Step 12: Process node C

Legend



Stack (top at right): J | G | F

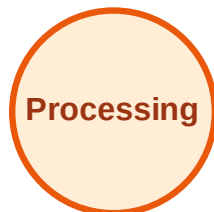
Visited: A, B, D, H, I



DFS Traversal

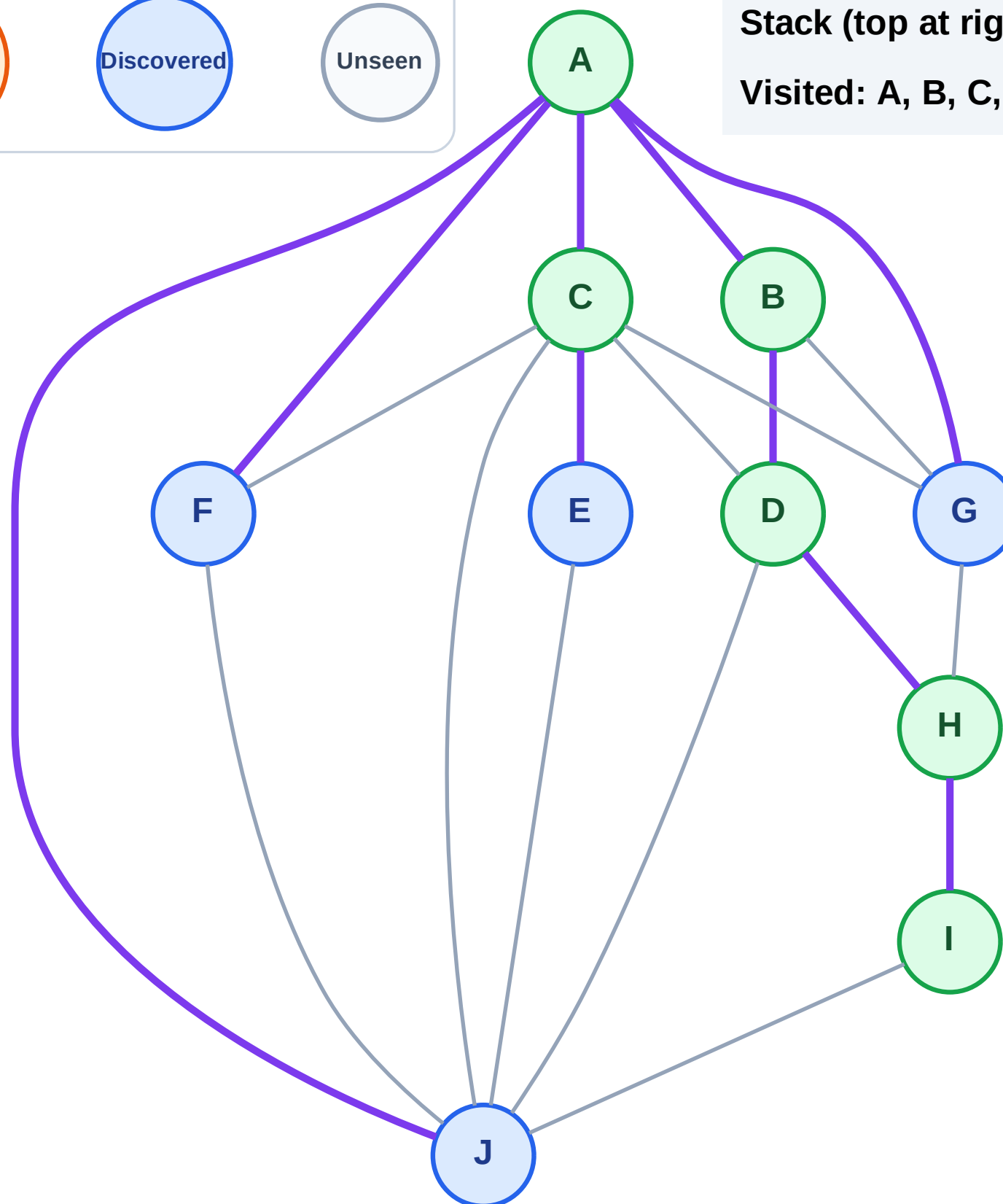
Step 13: Discover E from C

Legend



Stack (top at right): J | G | F | E

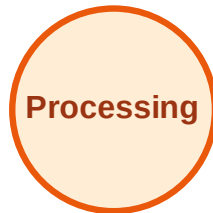
Visited: A, B, C, D, H, I



DFS Traversal

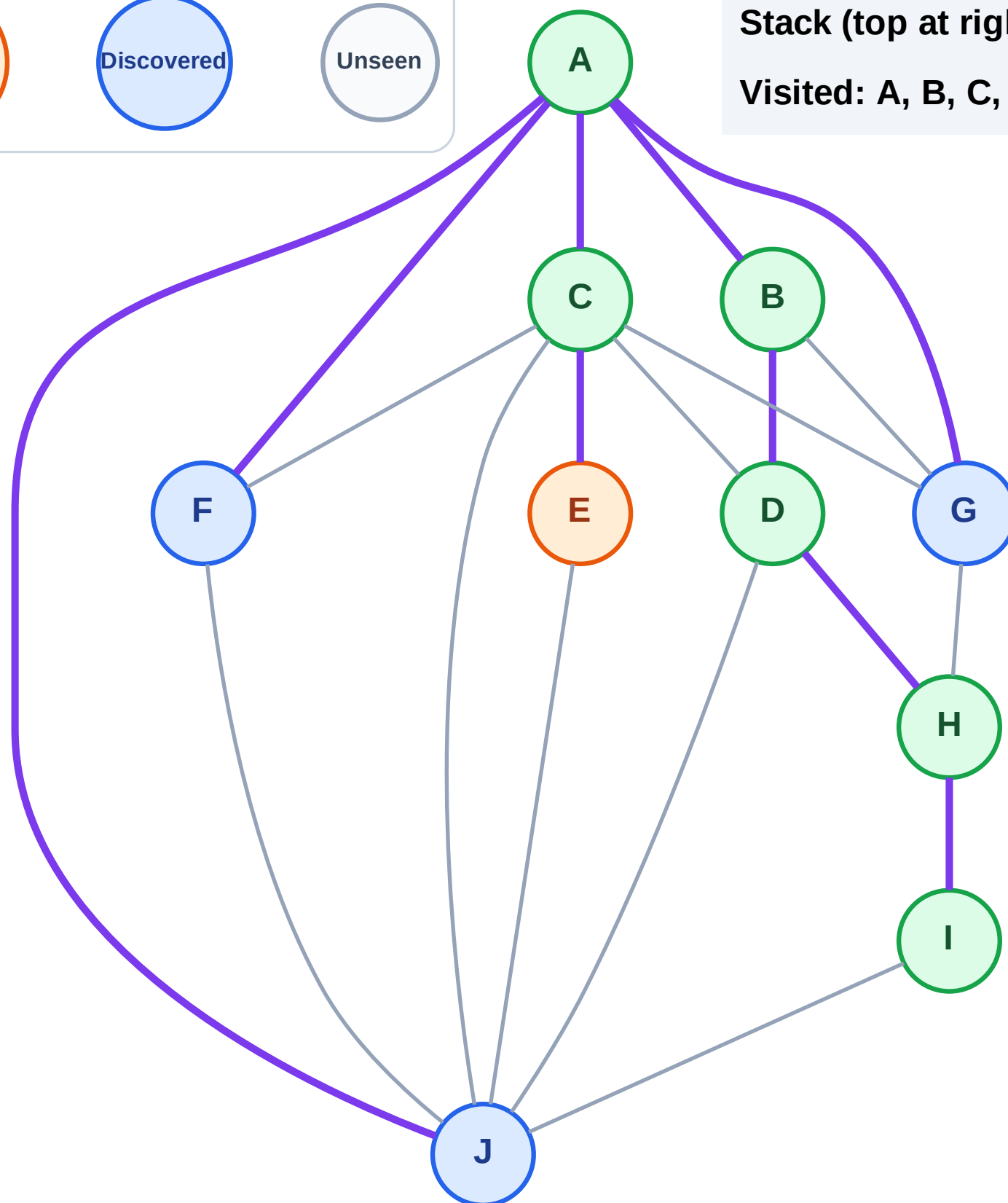
Step 14: Process node E

Legend



Stack (top at right): J | G | F

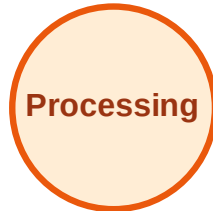
Visited: A, B, C, D, H, I



DFS Traversal

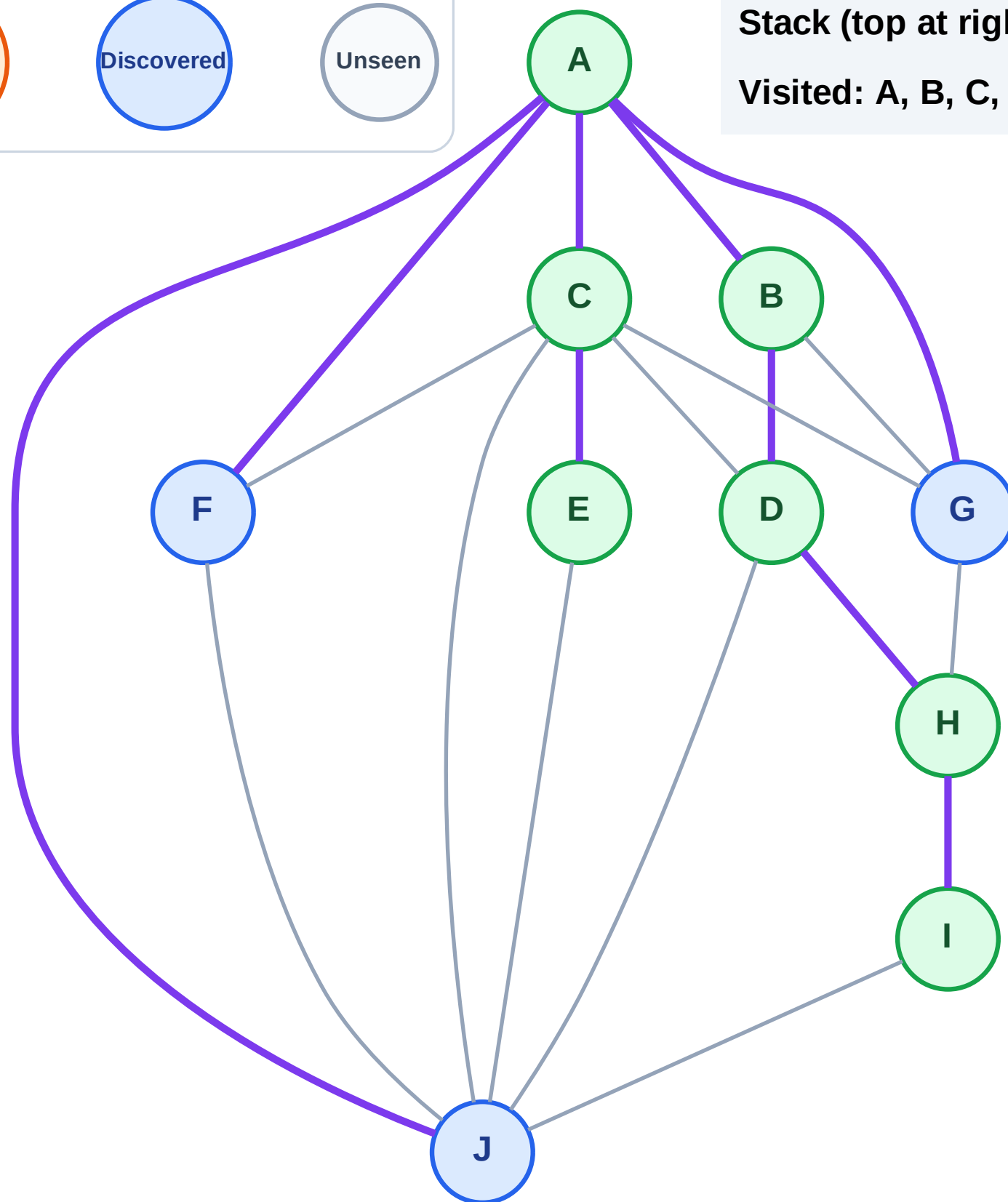
Step 15: No new neighbors from E

Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 16: Process node F

Legend

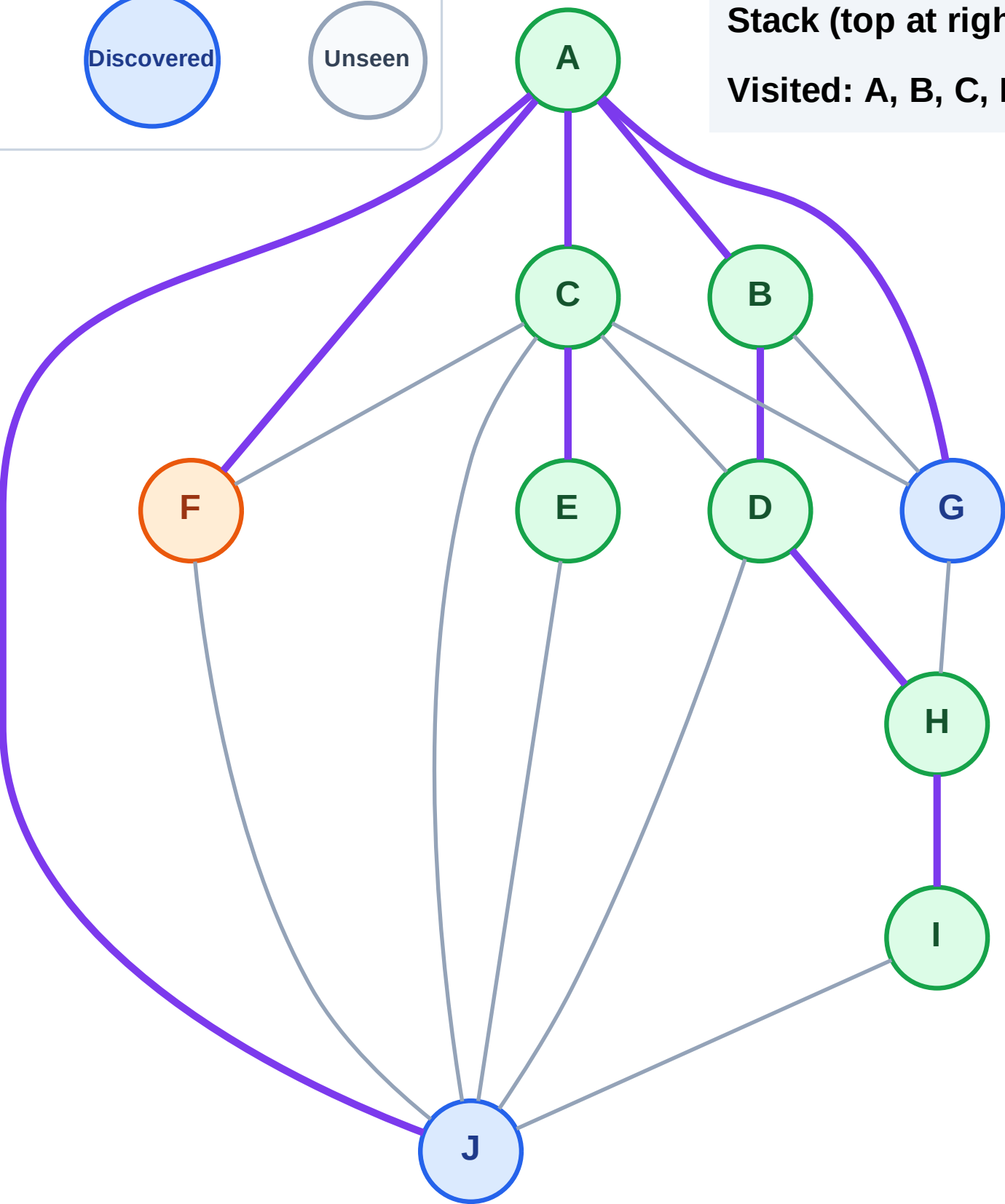
Complete

Processing

Discovered

Unseen

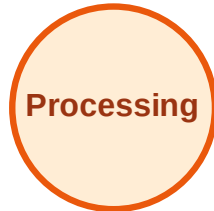
Stack (top at right): J | G
Visited: A, B, C, D, E, H, I



DFS Traversal

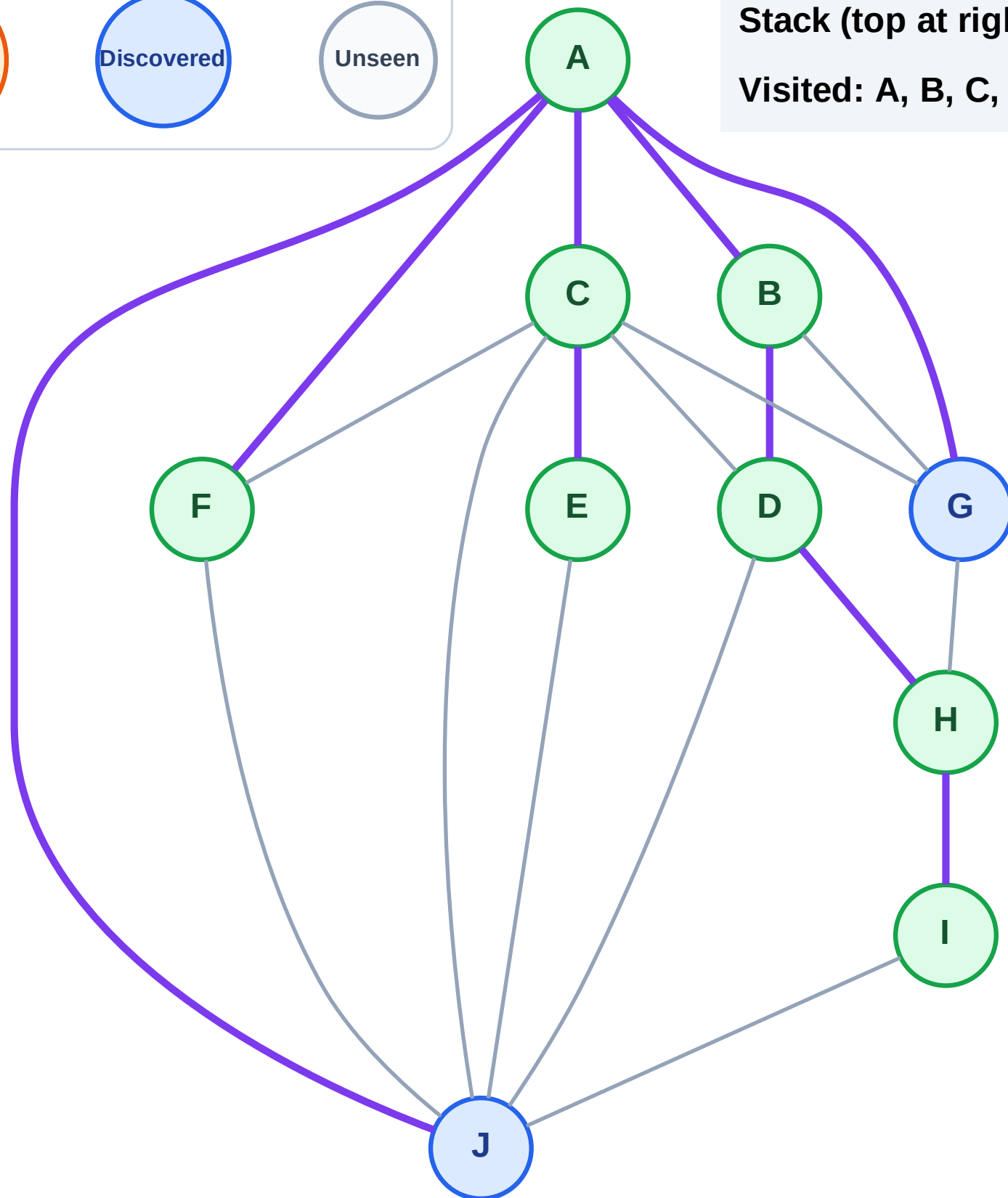
Step 17: No new neighbors from F

Legend



Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

Complete

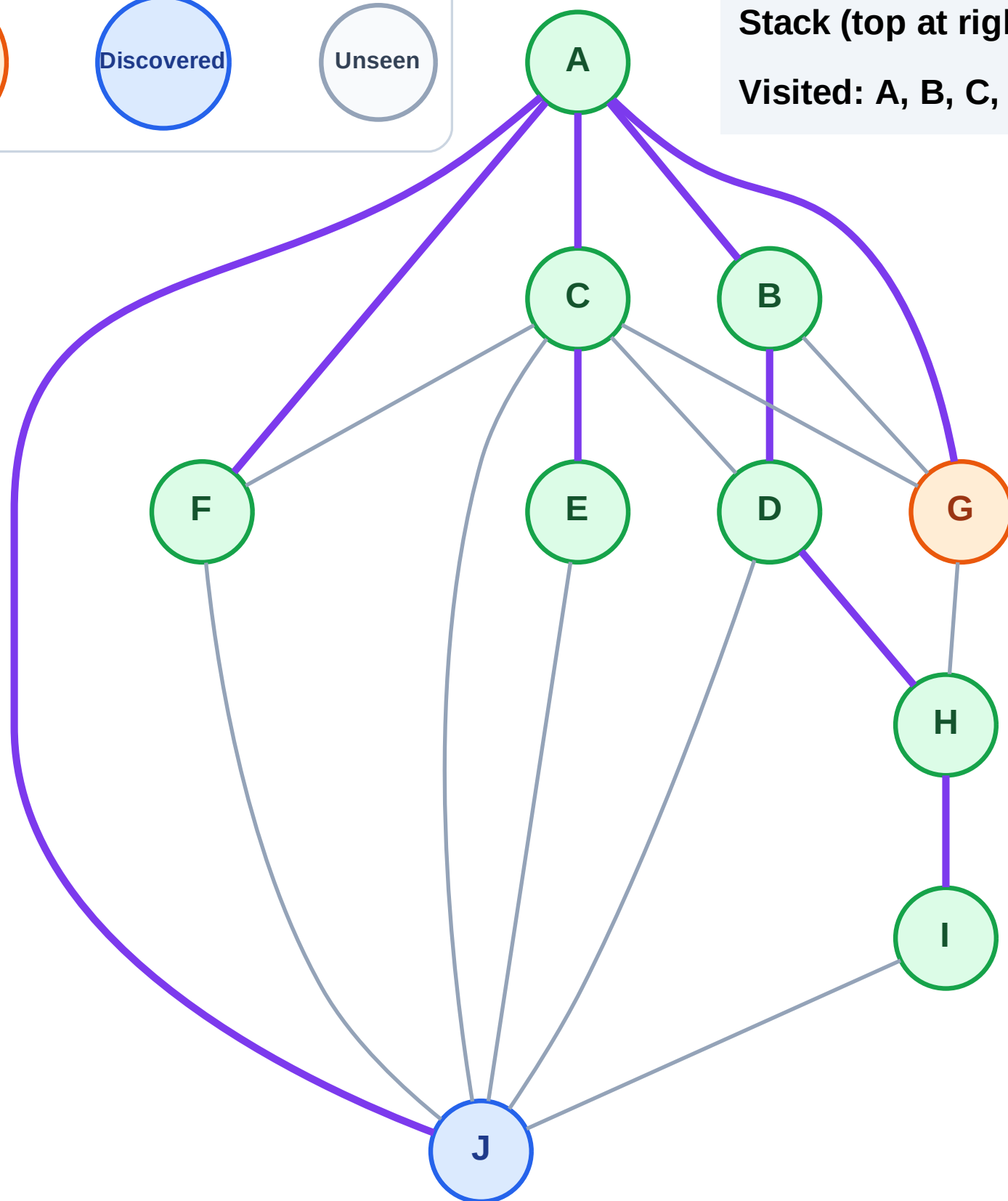
Processing

Discovered

Unseen

Stack (top at right): J

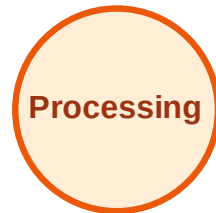
Visited: A, B, C, D, E, F, H, I



DFS Traversal

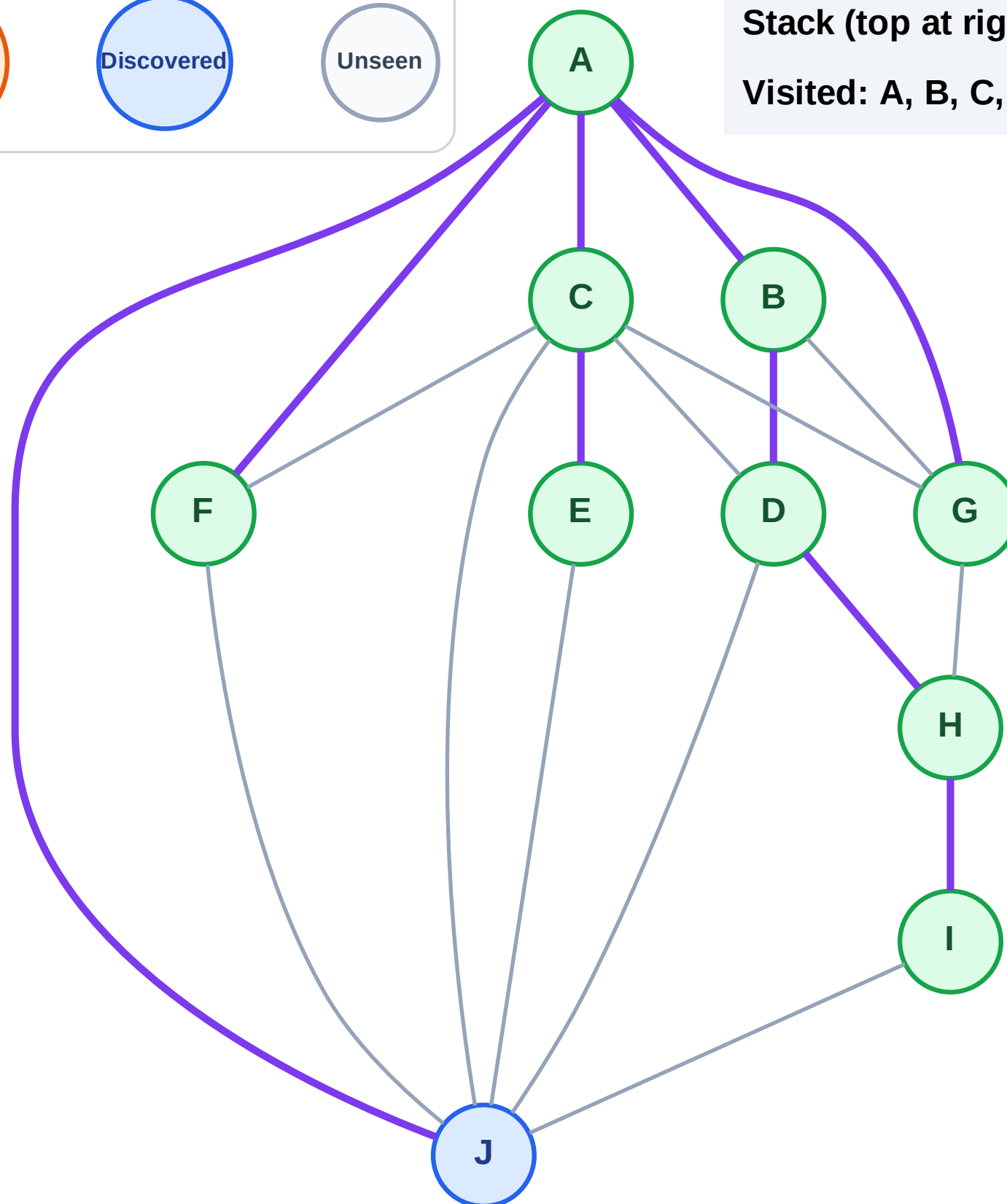
Step 19: No new neighbors from G

Legend



Stack (top at right): J

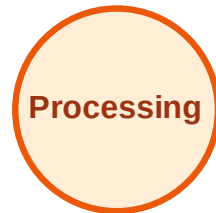
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

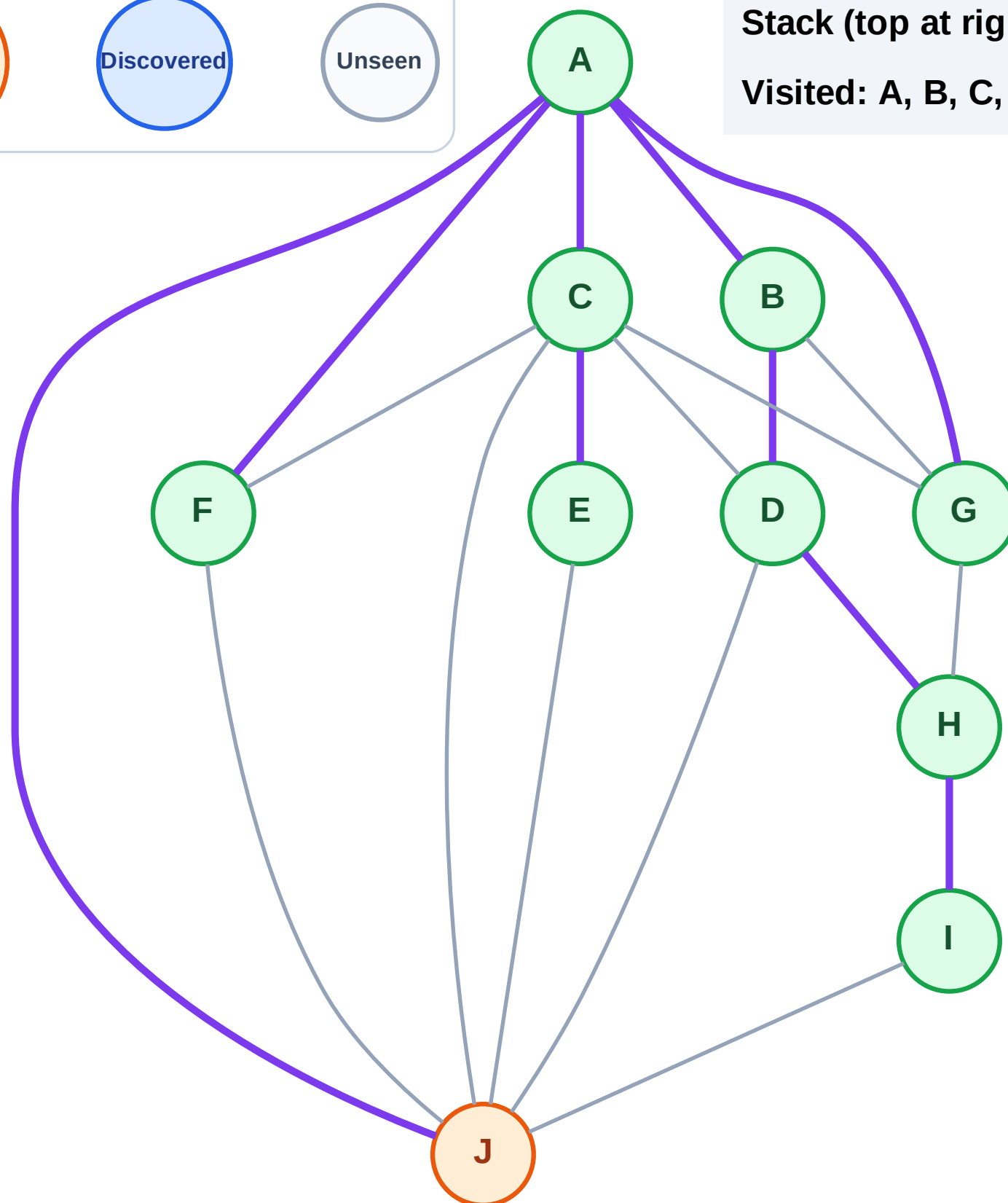
Step 20: Process node J

Legend



Stack (top at right): (empty)

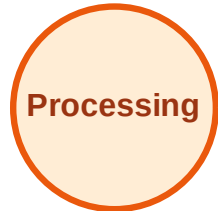
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

